

# Excitation-Aware On-Track System Identification at Full Scale: A Simulation Study

## Abstract

On-track system identification learns tire models from ordinary racing laps instead of dedicated manoeuvres. We study one such residual-learning pipeline at full vehicle scale, inside a simulator that publishes per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, it converges on a tire with less than half the vehicle's grip ceiling and most Pacejka coefficients resting on their bounds. The cause is a loss of excitation created by the pipeline's own virtual-data step: the friction the synthetic rollout can demand is a property of the rollout configuration and not of the tire, as a grid on a known tire confirms. Two further defects share that source. The objective the coefficients are fitted against depends on them only through one product, so the peak factor is not recoverable from the rollout at any excitation, and the loop that feeds each fit its own previous answer is not a contraction. We correct all three, and gate the controller interface on derived physical quantities rather than on coefficient bounds. On a static surface 33% below the one the tire prior was measured on, axle cornering stiffness is recovered to within 3.2% and the peak factor to 11%. On a surface losing half its grip under the moving vehicle, the corrected pipeline tracks the decay rate to 7% and completes three timed laps at unchanged lateral tracking error for 18% of lap time, where the inherited configuration holds a grip ceiling nearly four times the plant's and leaves the track.

**Keywords:** System identification, Tire models, Vehicle dynamics, Autonomous racing, Model predictive control, Persistent excitation

## 1 Introduction

Model-based controllers for autonomous racing derive their performance from the fidelity of the vehicle model they carry [1–3]. In the lateral channel that fidelity is dominated by the tire, simultaneously the physical bound on manoeuvrability and the least accessible component to measure [4, 5]. Classical characterisation takes the vehicle off the track for steady-state circular or ramp-steer manoeuvres, at a logistical cost a race weekend rarely affords; identifying on track removes the logistics but brings back transients, noise and a racing line that is a poor excitation signal, and nonlinear least squares applied directly to lap data is brittle in exactly this regime [6].

Dikici *et al.* [6] propose the hybrid remedy that is the starting point of this paper: a small residual network trained on 30 s of ordinary lap data predicts the one-step error of a nominal single-track model; the corrected model is rolled out through a *synthetic* quasi-steady steering sweep; Pacejka coefficients are fitted to that sweep; and each fit becomes the next nominal model. Because the sweep is generated rather than driven it can be made steady and noise-free, and the residual network only has to interpolate within its training distribution. On a 1:10 platform the procedure reports a  $3.3\times$  lower one-step RMSE than nonlinear least squares under noise [6].

## 1.1 Problem statement

This paper asks what the pipeline identifies when it is moved from a 3.7 kg, 0.32 m-wheelbase platform to a full-scale Formula Student vehicle: 269.6 kg, 1.53 m wheelbase,  $16^\circ$  of road-wheel lock, following a reference speed profile of 9.6–25.8 m/s. The question is identifiability, not parameter transfer, and two scale-dependent mechanisms, both latent at 1:10, break the estimate. First, excitation collapses inside the virtual-data step: the Pacejka fit never sees the recorded lap, only the synthetic rollout, whose reachable friction  $\mu_{\text{reach}} \approx v^2 \delta_{\text{sweep}} / (gL)$  is a property of the rollout configuration and not of the tire, and a rollout speed inherited from the scaled platform caps it at 0.43 here. Below the tire’s peak the Magic Formula degenerates to  $F_y \approx F_z(BCD)\alpha$ , only the product  $BCD$  is identifiable, and the bounded optimiser resolves the remaining freedom on a box edge [4, 7]. Second, lateral demand scales as  $v^2$ , so the tracking cost of a bad grip estimate grows quadratically with speed: one identified set with a peak friction coefficient of 0.40 tracked at 0.16 m of cross-track error at 11.1 m/s and 41.89 m at 27 m/s through the same controller (Section 7.1), and the scaled platform never reaches the speed at which that error becomes observable.

One difference works in our favour. Because the study runs in the CARLA simulator [8] through a purpose-built ROS 2 interface, the simulator’s own per-wheel telemetry (slip angle, normal load and lateral force) is available as ground truth, so an on-track identifier can be scored against the forces the plant actually generated.

## 1.2 Contributions

1. **An identifiability analysis of the virtual-data step** (Section 5.2): the closed-form reachable-friction bound above, and an *excitation-aware rollout* that selects the rollout speed from the measured lap and warns when the configuration cannot reach the prior’s peak. A grid over rollout speed, sweep amplitude and true peak factor on a known tire both verifies the bound and selects the rollout speed floor and sweep-amplitude ceiling the pipeline ships (Section 6.2). This corrects the baseline method [6] at any scale, and stays latent at 1:10 because a 2–4 m/s rollout is representative there and is not here.
2. **Two structural properties of the loop, and the corrections they imply** (Section 5.4): stage 3’s residual depends on  $B$ ,  $C$  and  $D$  only through their product, so the peak factor is not recoverable from the rollout *at any excitation* and must be measured on the recorded trace and held; and the loop is a positive feedback

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| path rather than a contraction, so stage 4 becomes a relaxed fixed-point iteration at $\beta = 0.20$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 093<br>094                                                                                                          |
| 3. <b>The corrections the bound implies</b> , on both sides of the fit (Sections 5.3 and 5.5): a sweep bounded at $1.5\times$ the 99th percentile of the observed steering, a Tikhonov target frozen at the configured prior, slip angles built from the <i>achieved</i> road-wheel angle, mass and wheelbase from measured static wheel loads, a friction warm start from the <i>curvature</i> of the axle force curve, and physics-informed dynamics residuals normalised to the state-increment units of the data term, from which raw force and moment residuals sit a factor $2.3 \times 10^5$ away.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 095<br>096<br>097<br>098<br>099<br>100<br>101                                                                       |
| 4. <b>Admissibility gates on derived physical quantities</b> at the model-to-controller boundary (Section 7): axle cornering stiffness, understeer sign and critical speed, and grip ceiling against the trajectory’s lateral demand. They replace per-coefficient box checks, which cannot detect a degenerate fit or a pairwise-oversteering axle set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 102<br>103<br>104<br>105                                                                                            |
| 5. <b>A simulation architecture that supplies the ground truth, and a closed-loop evaluation on two surfaces</b> (Sections 4.1–4.4 and 6). The first is static at 0.70, deliberately not the 1.05 the configured tire prior was measured on, so a pipeline unable to move off its prior scores badly. Over three timed laps the corrected pipeline identifies axle cornering stiffness to 3.2% front and 1.4% rear and tracks the peak factor to 11%, against an inherited configuration that rails $D$ on its 2.0 bound in every cycle and overstates stiffness by 118–132%; axle force RMSE is $12.1\times$ and $10.5\times$ lower and MPC lateral tracking error 32% lower, at unchanged solver cost. On the second the bridge drives the plant’s friction down under a moving vehicle on the linear-decay protocol of [9] until half the grip is gone. The corrected pipeline recovers the decay <i>rate</i> to 7%, which a single surface cannot test at any accuracy, and the level to 10.9–17.4% once one identification cycle of transport lag is accounted for. The falling peak factor then re-clamps the reference speed profile from 0.51 $g$ of peak lateral demand to 0.30 $g$ , and the vehicle finishes three laps at unchanged tracking error for 18% of lap time. The inherited configuration cannot make that trade, because a peak factor resting on a box constraint reports no change, and it leaves the track in lap 2. | 106<br>107<br>108<br>109<br>110<br>111<br>112<br>113<br>114<br>115<br>116<br>117<br>118<br>119<br>120<br>121<br>122 |
| All results are obtained in simulation, and Section 8 states which conclusions are expected to transfer to hardware.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 123<br>124<br>125                                                                                                   |
| <b>2 Related Work and Position</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 126<br>127                                                                                                          |
| <b>2.1 On-track and data-driven identification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 128<br>129                                                                                                          |
| The Magic Formula [4, 10] is the model identified here. Classical characterisation obtains its coefficients from steady-state manoeuvres in a controlled space [5, 6]: clean data at high logistical cost. Direct fitting to measured contact-patch forces needs a force-measuring hub or a simulator’s own wheel telemetry; here it is used only as a <i>reference</i> against which the on-track identifier is scored, never inside the identification loop.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 130<br>131<br>132<br>133<br>134<br>135                                                                              |
| On-track methods trade excitation quality for operational simplicity. Nonlinear least squares on lap data is the classical baseline and is brittle under noise [6]. The                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 136<br>137<br>138                                                                                                   |

recursive alternative is a dual or joint filter carrying states and parameters together [11, 12], and it is subject to the same excitation condition this paper is about: a filter run on a smooth racing line has as little information about the tire’s peak as a batch fit of the same data, and merely expresses that as a parameter covariance that never shrinks rather than as a coefficient on a bound. Dedicated friction estimators make the excitation dependence explicit [13–16], and Hsu *et al.* [17] is the closest prior art to the warm start of Section 5.5.2: they recover the friction limit below the peak from the *curvature* of the force response, read through the pneumatic trail in steering torque. Our estimator uses the same observation on the axle force curve reconstructed from the IMU and odometry, with no steering-torque sensor, and is then gated and pinned for the co-identification loop, which is a property of what that loop can identify (Section 5.4.1) rather than of the estimator. Gaussian-process approaches learn the mismatch of a nominal model [2, 18, 19] at a continuing computational cost; neural residuals replace the Gaussian process with a cheaper regressor at the cost of generalisation guarantees.

A changing surface is how that literature is usually stressed. LLA-MPC [9] selects from a bank of pre-built models instead of fitting one, and evaluates the choice on a friction coefficient decaying linearly in time and on abrupt drops between laps. We borrow its decay protocol rather than its controller (Section 6.1.1), because the question a model bank answers and the question this paper asks are different ones: a bank needs no excitation to switch and therefore cannot be starved of it, while an identifier that fits coefficients on line can be, which is the failure this paper locates. The decay is used here to test whether a fitted peak factor follows the plant, not to compare against a bank.

Deep Dynamics [20] is the closest work on the physics-constraint axis: it estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer clamping them into nominal ranges. Our admissibility gates share the motivation but differ in placement. We report as a substantive finding that per-coefficient box constraints are insufficient: the degenerate fit that motivated this work satisfied every box constraint while five of eight coefficients sat exactly on a bound, and a later fit satisfied every box constraint while being pairwise oversteering and open-loop unstable above 14.5 m/s. We therefore gate on derived quantities at the model-to-controller interface and treat a coefficient resting on a bound as a diagnostic of non-identifiability rather than a successful clamp. The hybrid residual-plus-virtual-data structure of [6] is the pipeline we extend; the idea it rests on, to query the learned model only where it interpolates and extract physically parameterised coefficients from it, is retained unchanged, and what we correct is the mechanism deciding *where* the learned model is queried.

## 2.2 Overlap with concurrent work

Wu *et al.* [21] extend the same baseline concurrently and independently, introducing a vision-based friction prior, an S4 structured state-space residual in place of the MLP, and Nelder–Mead for the iterative Pacejka extraction. The pipeline described here independently contains an S4D temporal residual [22, 23], a friction warm-start, and Nelder–Mead and differential-evolution solver options, so we claim none of the three as contributions. Two differences are technical rather than chronological. Our

friction prior is proprioceptive, read from the *curvature* of the axle force curve by a gated brush-model fit [15, 24] rather than from a camera or from utilised friction, which on a gentle lap measures 0.044 median on this vehicle and is only a lower bound (Section 5.5.2). And it is *pinned* rather than used as an initialisation, because Section 5.4.1 shows the rollout fit cannot separate the peak factor from the cornering stiffness at any excitation, so a prior on  $D$  the fit is free to move is overwritten within one cycle. Neither prior work is the fix: the degeneracy is located in the rollout excitation, not the initialisation, and to our knowledge the reachable-friction bound of Section 5.2 and its consequence for the virtual-data step have not been stated before.

### 2.3 Simulation platform and downstream consumer

CARLA [8] supplies the sensor and scenario infrastructure used here. Its native dynamics come from the underlying PhysX wheeled-vehicle model [25], a saturating friction-and-stiffness tire law (Section 4.2) rather than an empirical Magic Formula fit, which is why this study treats the simulator as a well-instrumented *plant* to be identified and not as a source of ground-truth Pacejka coefficients. R-CARLA [26] makes the dynamics model interchangeable, and Sagmeister *et al.* [27] find that a controller’s sensitivity to model fidelity is largest near the acceleration limits, where this study operates; CARLA has also hosted complete Formula Student Driverless stacks with ROS 2 interfaces [28]. Scaled platforms, F1TENTH [29] and the Forza-ETH Race Stack [3], provide the software context of the baseline; full-scale Formula Student Driverless systems [5] the vehicle class targeted here. The identified model is consumed by a nonlinear MPC path-tracking controller built on *acados* [30] with partial-condensing HPIPM [31] [1, 2], tracking a minimum-curvature raceline [32]. We claim no contribution to the controller; it appears only as the consumer that defines what an admissible identification is (Section 7).

## 3 Identification Problem

### 3.1 Model structure

The identification model is the dynamic single-track model of the baseline, with mass  $m$ , yaw inertia  $I_z$  and centre-of-gravity distances  $l_f$ ,  $l_r$ ,  $L = l_f + l_r$ . Longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \frac{1}{m} (F_{y,r} + F_{y,f} \cos \delta - m v_x \omega), \quad (1)$$

$$\dot{\omega} = \frac{1}{I_z} (F_{y,f} l_f \cos \delta - F_{y,r} l_r), \quad (2)$$

with slip angles

$$\alpha_f = \delta - \arctan\left(\frac{v_y + l_f \omega}{v_x}\right), \quad \alpha_r = -\arctan\left(\frac{v_y - l_r \omega}{v_x}\right). \quad (3)$$

Axle forces follow the lateral Magic Formula

$$F_y = F_z D \sin(C \arctan(B\alpha - E(B\alpha - \arctan B\alpha))), \quad (4)$$

with per-axle coefficients  $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$ . The identification state and input are  $\mathbf{x} = [v_y, \omega]$  and  $\mathbf{u} = [v_x, \delta]$ , and the nominal one-step map is  $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$ . The baseline discretises Eqs. (1)–(2) by explicit Euler at  $T_s$ ; the integrator is selectable and the shipped configuration uses RK4, because the full-scale vehicle’s lateral mode is stiffer relative to  $T_s$  than the scaled platform’s. The inherited arm of Section 6 keeps Euler.

### 3.2 Objective and derived quantities

The objective is to recover  $\Phi_p$  from a single 30 s segment of ordinary lap data, without dedicated manoeuvres, such that the resulting model is both predictive and *admissible* for the controller that consumes it. Admissibility is expressed on derived quantities rather than on coefficients. The linearised axle cornering stiffness is the slope of Eq. (4) at zero slip,

$$C_i = F_{z,i} B_i C_i^{\text{MF}} D_i, \quad i \in \{f, r\}, \quad (5)$$

(writing  $C^{\text{MF}}$  for the shape factor to avoid a clash), and the grip ceiling implied by an identified set is

$$a_y^{\max} = \frac{D_f F_{z,f} + D_r F_{z,r}}{m}. \quad (6)$$

The linear model built from Eq. (5) is oversteering when  $l_f C_f > l_r C_r$  and is then open-loop unstable above the critical speed  $v_{\text{crit}} = L \sqrt{C_f C_r / (m(l_f C_f - l_r C_r))}$  [33].

### 3.3 Practical identifiability under limited excitation

Two properties of Eq. (4) are used throughout. Its slope at zero slip is  $F_z BCD$ , so the cornering stiffness depends on the coefficients only through that product, and  $E$  drops out. And if the data never approaches the peak, Eq. (4) linearises to  $F_y \approx F_z (BCD)\alpha$ , leaving  $B$ ,  $C$  and  $D$  mutually unidentifiable. Given data that does reach the peak, Eq. (4) is *structurally* identifiable [34]; what fails here is *practical* identifiability [35], the standard persistent-excitation condition [7, 36] written on this model, and we claim neither result as novel. Our claim, demonstrated in Section 5.2, is that the baseline pipeline’s virtual-data step can silently violate that condition, and that a bounded optimiser then resolves the remaining freedom by sliding onto a box edge. The violation surfaces as a converged fit rather than a failure.

### 3.4 Baseline pipeline

The baseline repeats four stages for  $N_{\text{it}}$  iterations [6]. A small network is trained on the recorded lap to predict the nominal model’s one-step error from  $[v_x, v_y, \omega, \delta]_k$  (stage 1); the corrected model is integrated open-loop at constant speed while the steering angle is ramped from 0 to  $\delta_{\text{sweep}}$ , producing a synthetic quasi-steady dataset (stage 2); assuming  $\dot{v}_y = \dot{\omega} = 0$  on that data, axle forces are recovered from the yaw balance,

$$F_{y,r} = \frac{m l_f v_x \omega}{L}, \quad F_{y,f} = \frac{m l_r v_x \omega}{L \cos \delta}, \quad (7)$$

and  $\Phi_p$  is fitted to  $(\alpha_i, F_{y,i})$  by bounded least squares (stage 3); the fitted  $\Phi_p$  then becomes the next iteration’s nominal model (stage 4). Algorithm 1 gives the corrected cycle in full.

The rest of this paper follows from one property of that structure: stage 3 never sees the recorded lap, only the output of stage 2. The recorded lap enters only through the residual network, so the excitation available to the fit is set by the rollout configuration and not by how the vehicle was driven.

## 4 Simulation Architecture and Data Pipeline

The pipeline is exercised end to end inside a single architecture: a Formula Student vehicle model in CARLA acting as the plant, a C++ ROS 2 communication layer that drives it and republishes its state, and an identification stack that consumes that state. CARLA [8] was selected because it exposes per-wheel slip angle, normal load and tire force rather than chassis states alone, and because its synchronous mode with a fixed physics step (0.033s here) stamps every sample on the simulation clock; those two properties are what make the identifiability claims of Section 5 checkable rather than arguable [37].

### 4.1 Plant: Formula AI vehicle in CARLA

The plant is the driverless Formula Student-class vehicle model, instantiated in CARLA (server 0.9.16, Unreal Engine 4.26) on the **silverstone** map: a four-wheeled, front-steered, all-wheel-drive rigid-body vehicle whose lateral forces are generated by the PhysX wheeled-vehicle tire model [25] under a configured friction coefficient and lateral stiffness, not by a Magic Formula. The simulator therefore has no “true” Pacejka coefficients to recover; instead the lateral forces PhysX computes at each wheel are taken as ground truth, and the identified Magic Formula is validated against those forces.

### 4.2 The plant’s tire curve, in closed form

The identification target is known here in closed form. PhysX computes the lateral force of each wheel with `PxVehicleComputeTireForceDefault` [25]; at zero longitudinal slip and zero camber that expression collapses to

$$F_y(\alpha) = \text{sgn}(\alpha) \mu F_z S_1(K), \quad K = \frac{C_{\text{lat}} |\tan \alpha|}{\mu F_z}, \quad (8)$$

with  $C_{\text{lat}} = F_z^{\text{rest}} k_y S_1(3\bar{F}_z/k_x)$ ,  $\bar{F}_z = F_z/F_z^{\text{rest}}$  the normalised load,  $(k_y, k_x)$  the wheel’s lateral-stiffness parameters, and  $S_1(K) = \min(1, K - K^2/3 + K^3/27)$  the engine’s smoothing function.

Two properties of Eq. (8) matter downstream. It saturates at  $\mu F_z$ , first reached at  $K = 3$ , and it never falls off past that peak, so a Magic Formula can match it up to the peak and no further:  $E$  and the post-peak decay exist to describe a falling branch this curve does not have. Its scale is set entirely by  $\mu$  and  $C_{\text{lat}}$ , both of which the simulator publishes, so the curve of Table 2 is read off the plant rather than fitted to it.

**Table 1** Vehicle configuration and measured plant properties.

| Quantity                               | Value           | Unit              |
|----------------------------------------|-----------------|-------------------|
| Body mass (configured)                 | 240.0           | kg                |
| Static wheel load (total)              | 2644.5          | N                 |
| Total (kerb) mass $m^a$                | 269.6           | kg                |
| Front/rear static split <sup>b</sup>   | 51.2 / 48.8     | %                 |
| Wheelbase $L$                          | 1.5334          | m                 |
| $l_f / l_r$                            | 0.7381 / 0.7953 | m                 |
| Yaw inertia $I_z^c$                    | 51.1            | kg m <sup>2</sup> |
| Max road-wheel angle                   | 16.0            | °                 |
| Configured wheel friction <sup>d</sup> | 1.0             | —                 |
| Road-surface factor                    | 0.70            | —                 |
| Effective friction <sup>e</sup>        | 0.70            | —                 |
| Realised peak axle friction            | 0.71            | —                 |

<sup>a</sup>Total static wheel load divided by  $g$ ; this is the mass every equation in this paper uses, not the configured body mass. <sup>b</sup>The per-axle loads used elsewhere in this paper (Table 2, Section 6),  $F_{z,f} = 1373$  N and  $F_{z,r} = 1274$  N, follow  $m, l_f, l_r$  through the static load-transfer split (51.9/48.1 %) rather than this directly measured one; Appendix A (A2) reports the 1.4 % disagreement between the two splits. <sup>c</sup>Assumed; see Appendix A. <sup>d</sup>A scenario parameter, not a vehicle property. <sup>e</sup>Configured wheel friction multiplied by the road surface's own coefficient, as reported per wheel. It is the effective value, not the configured one, that the identifier is scored against (Section 5.5.2).

**Table 2** The plant's lateral tire curve: the parameters of Eq. (8), read from the simulator's own wheel configuration and telemetry.

| Quantity                                     | Value                      | Unit   |
|----------------------------------------------|----------------------------|--------|
| Effective friction $\mu$                     | 0.70                       | —      |
| Lateral stiffness $k_y$                      | 17.0                       | /rad   |
| Max normalised load $k_x$                    | 2.0                        | —      |
| Initial slope $C_{lat}$ at $\bar{F}_z = 1$   | 14.875 $F_z^{\text{rest}}$ | N/rad  |
| front axle, $F_{z,f} = 1373$ N               | 20.4                       | kN/rad |
| rear axle, $F_{z,r} = 1274$ N                | 19.0                       | kN/rad |
| Peak reached at, $\bar{F}_z = 1, \mu = 0.70$ | 8.0                        | °      |

The initial slope is set by the *rest* load and by the normalised load through  $S_1$ , not by  $F_z$  alone, and falls monotonically with  $\bar{F}_z$ . Every axle slope quoted in this paper is taken at the static axle loads listed here; the closed-form values in this table are the  $\bar{F}_z = 1$  ones, and a slope fitted to telemetry from a run that loads the outside wheels beyond the rest load comes out below them: 17.4 kN/rad front on the  $\mu = 0.70$  run of Section 6.4, against 20.4 kN/rad here.

The peak angle follows from  $K = 3$  in Eq. (8) and therefore moves with  $\mu$ :  $8.0^\circ$  at the effective  $\mu = 0.70$  of the runs reported here,  $12.0^\circ$  at the  $\mu = 1.05$  of the prior-fitting run.

That the closed form is the right one was checked rather than assumed: on 1804 ground-truth wheel samples from a live lap ( $P_{99}(|\alpha|) = 3.66^\circ$ ), each predicted from its own measured  $\alpha$ ,  $F_z$  and  $\mu$ , Eq. (8) returns  $R^2 = 0.99$  at 16.1 N front and 14.7 N rear RMSE. Section 6.4 uses this curve to separate the error of the identified fit from the error of representing Eq. (8) in the form Eq. (4) that the controller requires.

### 4.3 The simulator bridge: interfaces and telemetry

SIMBRIDGE is a C++ ROS 2 lifecycle node linked directly against CARLA's client library (Figure D2, Appendix D). It spawns and configures the ego vehicle, drives

its inputs over a single `ackermann_msgs/AckermannDriveStamped` on `/drive`, and republishes simulator state as standard ROS 2 topics; all identification traffic crosses this interface. On the feedback path, one property is specific to this architecture and the rest of this paper rests on it: the simulator’s per-wheel slip angle, normal load and lateral force are republished on the same simulation clock as the odometry the identifier consumes, so an identified tire model can be scored against the forces the plant actually generated rather than a second model. Appendix D lists the full interface and the body-frame and sign conventions at the simulator–ROS 2 boundary.

#### 4.4 Data acquisition and preprocessing

A model-free pure-pursuit controller drives the vehicle around the reference raceline; it requires no tire knowledge beyond the wheelbase and is therefore a valid bootstrap, as in the baseline [3, 6]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer by a 50 Hz timer for 30 s, gated on  $v_x > 1$  m/s. The timer runs on the simulation clock, which advances only at the server’s 0.033 s step, so the achieved acquisition rate is the step rate.

Two mismatches follow. The acquisition timer asks for a sample every 0.020 s, 65 % more often than the server produces one, so a fraction of consecutive buffer entries are repeated states and the one-step residual target carries a zero-order-hold artefact on those pairs. Separately, the one-step model of Section 3.1 is discretised at  $T_s = 0.035$  s, 6 % longer than the physics step.  $T_s$  is empirical and is carried because the identification proved insensitive to it: at 0.020 s and 0.035 s the corrected configuration returns the *same* last accepted coefficient set to four decimal places, and the inherited configuration keeps  $D$  on its 2.0 bound at both, so the box-edge signature is not an artefact of the mismatch. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring  $(v_y, \omega, \delta) \rightarrow (-v_y, -\omega, -\delta)$  at unchanged  $v_x$ .

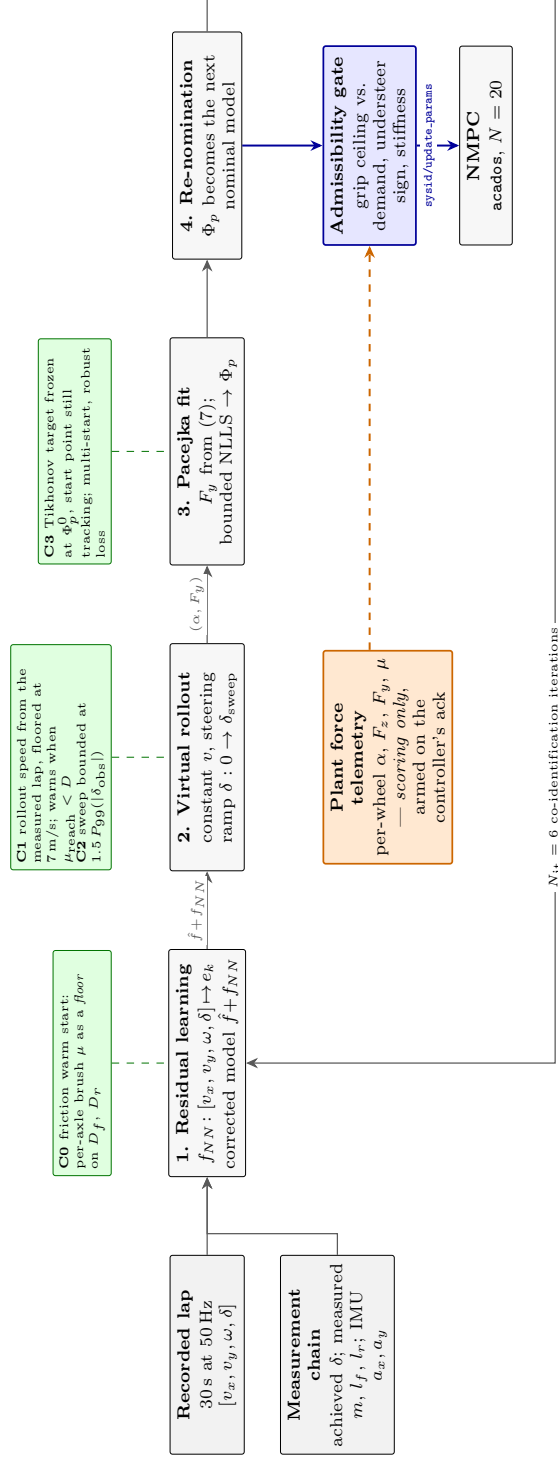
Only the deployment stage leaves this architecture. Validation draws on `feedback/tire_forces` and `feedback/imu`, neither of which enters the identification loop, so the estimator is scored against a channel it did not observe and Section 6 can attribute a closed-loop difference to the identified model rather than to the interface carrying it.

## 5 Identification Methodology

Figure 1 summarises the loop and Algorithm 1 states it in the form needed to reimplement it; lines marked ▷ are this paper’s changes to the inherited pipeline. Sections 5.2–5.3 carry the central methodological contribution.

### 5.1 Residual model and objective

The residual carries whatever the nominal model Eqs. (1)–(4) does not represent. The default is the baseline multilayer perceptron: 4 inputs, one hidden layer of 8 units, 2 outputs, LeakyReLU, 58 parameters, full-batch Adam [38] at  $5 \times 10^{-4}$  on an MSE objective, in PyTorch [39]. The architecture is selectable; the two arms of Section 6 use `baseline` and `s4`, the latter an S4D diagonal state-space residual [22]



**Fig. 1** The identification pipeline. Recorded lap data trains the residual network, the corrected model is rolled out through a synthetic steering sweep, Pacejka coefficients are fitted to that sweep, and the fit becomes the next nominal model under relaxation. Green marks this paper's changes: the friction warm start (Section 5.5.2), the excitation-aware rollout speed and extrapolation-bounded sweep (Sections 5.2-5.3), the frozen regularisation target of Eq. (12) and the relaxed update of Eq. (14). Blue is the gated controller interface (Section 7); orange is the plant's per-wheel force telemetry, which scores the result and never enters the loop.

|                                                                                                                                                                                           |     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| in: buffer $\mathcal{D} = \{v_x, v_y, \omega, \delta^{\text{ach}}\}_{1:K}$ (30 s), prior $\Phi_p^0$ , bounds $[\Phi_p^-, \Phi_p^+]$                                                       | 461 |
| out: $\Phi_p$ , or UNIDENTIFIED                                                                                                                                                           | 462 |
| 1: $\delta \leftarrow$ achieved road-wheel angle, command on staleness                                                                                                                    | 463 |
| 2: $m, l_f, l_r \leftarrow$ measured static wheel loads                                                                                                                                   | 464 |
| 3: $\hat{\mu}_f \leftarrow \text{BRUSHFIT}(\mathcal{D})$ , front axle, released only if gated                                                                                             | 465 |
| 4: <b>if</b> released <b>then</b> $D_f^0, D_r^0 \leftarrow \hat{\mu}_f$ ; $D$ held for the whole loop                                                                                     | 466 |
| 5: $\Phi_p^{\text{tik}} \leftarrow \Phi_p^0$ (frozen for all iterations)                                                                                                                  | 467 |
| 6: $v \leftarrow \text{clip}(\bar{v}_x(\mathcal{D}), v_{\min}, \infty)$                                                                                                                   | 468 |
| 7: $\delta_{\text{sweep}} \leftarrow \text{clip}(1.5 P_{99}( \delta ), \delta_{\min}, \delta_{\max})$                                                                                     | 469 |
| 8: $\mu_{\text{reach}} \leftarrow v^2 \delta_{\text{sweep}} / (gL)$ ; <b>warn</b> <b>if</b> $\mu_{\text{reach}} < \max_a D_a^0$                                                           | 470 |
| 9: $\Phi_p \leftarrow \Phi_p^0$ ; <b>for</b> $i = 1$ <b>to</b> $N_{\text{it}}$ <b>do</b>                                                                                                  | 471 |
| 10: $f_{NN} \leftarrow \text{TRAINRESIDUAL}(\mathcal{D}, \Phi_p)$ ; $\mathcal{S} \leftarrow \text{ROLLOUT}(\Phi_p, f_{NN}, v, \delta_{\text{sweep}})$                                     | 472 |
| 11: $\tilde{\Phi}_p \leftarrow \arg \min_{B,C,E} \ F_y(\Phi_p; \mathcal{S}) - \hat{F}_y(\mathcal{S})\ ^2 + \lambda \ \Phi_p - \Phi_p^{\text{tik}}\ ^2$ , $D$ held                         | 473 |
| 12: $\Phi_p \leftarrow (1 - \beta) \Phi_p + \beta \tilde{\Phi}_p$ ( $\beta = 0.20$ )                                                                                                      | 474 |
| 13: <b>if</b> any $\Phi_p$ component on a bound <b>then</b> report the degeneracy                                                                                                         | 475 |
| 14: <b>if not</b> $\text{GATES}(\Phi_p)$ <b>then return</b> UNIDENTIFIED                                                                                                                  | 476 |
| 15: submit $\Phi_p$ ; keep the previous model until acknowledged                                                                                                                          | 477 |
| : $\text{GATES}$ : axle stiffness sign and magnitude; $l_f C_f < l_r C_r$ and $v_{\text{crit}}$ outside the speed range; grip ceiling $\geq$ the trajectory's lateral demand (Section 7). | 478 |
| Algorithm 1: Corrected on-track identification cycle. $\triangleright$ marks a change to the inherited pipeline [6].                                                                      | 479 |

over a sliding window of  $W = 20$  samples (0.7 s), so the residual can carry tire-relaxation and yaw-damping dynamics a per-sample map cannot. We claim neither that architecture, which is also in concurrent work [21], nor the  $8.7\times$  reduction in its training cost, an exact algebraic rewrite that changes no identified coefficient; both are in the supplementary material, and the identifiability result below holds for either architecture.

The input space of the **s4** arm is augmented with the axle slip angles of Eq. (3): since the tire law Eq. (4) is a function of  $(\alpha, F_z)$  and of nothing else, supplying  $\alpha_f, \alpha_r$  directly spares the network a relation the model already states. The augmentation is also available on the MLP, at 16 extra weights ( $58 \rightarrow 74$  parameters), and is off in the **baseline** arm. The optional physics-informed objective

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_s \mathcal{L}_{\text{steady}} + \lambda_m \mathcal{L}_{\text{sym}} + \lambda_g \mathcal{L}_{\text{smooth}}, \quad (9)$$

with  $\lambda_s = 0.1$ ,  $\lambda_m = 0.05$ ,  $\lambda_g = 10^{-4}$ , adds three penalties to the one-step MSE [20, 40]:  $\mathcal{L}_{\text{steady}}$  keeps the two residual channels jointly consistent with Eqs. (1)–(2) on a quasi-steady mask, so they cannot imply an axle force pair no tire could produce (stage 3 fits that force pair, not the state prediction);  $\mathcal{L}_{\text{sym}}$  enforces the odd symmetry  $e(-\mathbf{z}) = -e(\mathbf{z})$  of a laterally symmetric vehicle;  $\mathcal{L}_{\text{smooth}}$  is a soft Lipschitz bound.

One property of Eq. (9) is not a hyperparameter.  $\mathcal{L}_{\text{data}}$  is a mean square over a one-step increment while the two balance residuals are a force and a moment, four to six orders of magnitude apart, so  $\lambda_s$  is not a relative weight at all: measured at initialisation on a 30 s buffer of this vehicle, the unnormalised steady term contributes 99.9996 % of  $\mathcal{L}$ , a ratio of  $2.3 \times 10^5$ , and such a residual fits the nominal model's own force imbalance rather than the data it was given. Each residual is therefore expressed

507 as the state increment it implies, the unit the network’s output is in,

$$508 \quad r_{\text{lat}} = \frac{T_s}{m} \left( m\dot{v}_y + mv_x\omega - (F_{y,r} + F_{y,f} \cos \delta) \right), \quad (10)$$

511 and  $r_{\text{yaw}}$  likewise with  $T_s/I_z$ . The shares become 98.53% data and 1.47% steady,  
 512 recovering the meaning Eq. (9) gives  $\lambda_s$ . The weights in Eq. (9) are relative weights  
 513 only in these units.

## 514 5.2 Excitation-aware virtual rollout

517 Consider stage 2 of Section 3.4: a constant-speed rollout at  $v$  with the steering ramped  
 518 to  $\delta_{\text{sweep}}$ . In quasi-steady cornering the lateral acceleration a single-track vehicle devel-  
 519 ops kinematically is  $a_y = v^2\kappa$  with  $\kappa \approx \delta/L$ , so the largest friction coefficient the  
 520 rollout can *demand* of the tire is

$$521 \quad \mu_{\text{reach}} \approx \frac{v^2 \delta_{\text{sweep}}}{gL} \quad (11)$$

525 Eq. (11) contains no tire parameter: it is a property of the rollout configuration alone,  
 526 and an *upper* bound rather than an equality, since the steady-state curvature a com-  
 527 pliant vehicle holds is smaller than the Ackermann  $\delta/L$ . Reading it as a ceiling is all  
 528 the argument needs: if even the ceiling sits below  $D$ , the synthetic data never leaves  
 529 the linear ramp of Eq. (4), only the product  $BCD$  is identifiable (Section 3.3), and  
 530 the bounded optimiser resolves the residual freedom on a box edge.

531 The baseline implementation selects the rollout speed as  $v = \text{clip}(\bar{v}_x, 2, 4)$  m/s,  
 532 representative of a 1:10 platform. On our vehicle, with  $\delta_{\text{sweep}} = 0.4$  rad and  $L =$   
 533  $1.5334$  m, Eq. (11) gives  $\mu_{\text{reach}} = 0.43$  at 4 m/s: the rollout cannot ask the tire for  
 534 more than  $0.43g$  whatever the tire is. At 1:10 scale, with  $L \approx 0.32$  m, the same speed  
 535 yields  $\mu_{\text{reach}} \approx 2.0$  and the pathology is invisible. The rollout speed is therefore now  
 536 taken from the speed the vehicle was actually driven at, clamped below at 15 m/s,  
 537 and the pipeline warns whenever Eq. (11) evaluates below the prior’s  $D$ , turning a  
 538 silent degeneracy into a startup diagnostic. The same bound sets a data-collection  
 539 requirement: identification data should be gathered at  $v \geq \sqrt{gLD/\delta_{\text{sweep}}}$ .

## 541 5.3 Bounded extrapolation and frozen-prior regularisation

542 Raising the rollout speed exposes a second, opposite failure: the rollout queries the  
 543 residual network at steering angles it has never seen, the observed  $|\delta|$  on a pure-pursuit  
 544 lap staying below  $\approx 0.06$  rad while the sweep ran to 0.4 rad. The fit reads the network’s  
 545 extrapolation as additional grip, and because each iteration’s answer became both the  
 546 next iteration’s nominal model *and* its regularisation target, the error compounds:  $D$   
 547 walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, a claimed 1.7  $g$   
 548 on a vehicle measuring 1.0  $g$  in steady state.

549 Two changes close the loop. First, an extrapolation bound: the sweep terminates  
 550 at  $\delta_{\text{sweep}} = \text{clip}(\lambda_e P_{99}(|\delta_{\text{obs}}|), \delta_{\text{min}}, \delta_{\text{max}})$  with  $\lambda_e = 1.5$ ,  $\delta_{\text{min}} = 0.05$  rad and  $\delta_{\text{max}} =$   
 551  
 552

0.54 rad, so the network is never queried more than 50 % beyond the steering range it was trained on. Both  $\delta_{\max}$  and the inherited 0.4 rad lie above the plant's 0.279 rad mechanical lock deliberately: the sweep is a *virtual* rollout, and clamping it to the lock would cap  $\mu_{\text{reach}}$  a second time, by the actuator instead of the rollout speed (Appendix A, A6). Second, a frozen prior: the Tikhonov target is held at the configured  $\Phi_p^0$  for all iterations while the optimiser's *start point* continues to track the previous iteration, giving the per-axle objective

$$\min_{\phi} \sum_k \rho(r_k(\phi)) + \sum_j w_j (\phi_j - \phi_j^0)^2, \quad (12)$$

with  $r_k$  the force residual,  $\rho$  a soft- $\ell_1$  robust loss and  $w_j$  scaled to the data residual budget by a single dimensionless weight  $\lambda = 0.05$ . The regulariser exists to hold the coefficients the manoeuvre does not excite,  $E$  above all, at their nominal value instead of on a box edge.

## 5.4 What the rollout fit identifies, and why the loop must be damped

### 5.4.1 The rollout fit sees only the product $BCD$

In the virtual-data step the low-slip degeneracy of Section 3.3 is stronger, and structural rather than excitation-dependent. Stage 3 does not observe a force; it reconstructs one from the rollout's own quasi-steady yaw balance Eq. (7), which on the synthetic trajectory reduces to

$$\frac{F_{y,f}}{F_{z,f}} = \frac{v_x \omega}{g \cos \delta}, \quad \frac{F_{y,r}}{F_{z,r}} = \frac{v_x \omega}{g}, \quad (13)$$

so the only quantity Eq. (12) ever compares against is the lateral acceleration the rollout realises, which the *previous* coefficient set produced through its slope  $F_z BCD$ . The fit's residual therefore depends on  $(B, C, D)$  only through their product, and  $D$  is the free coordinate of the hyperbola  $BCD = \text{const}$ . Replayed on five identification buffers of a live  $\mu = 1.05$  run, the product is recovered tightly,  $BCDF_{z,f}$  over 18.9–21.1 kN/rad against the 20.4 kN/rad the plant carries at  $\bar{F}_z = 1$  (Table 2; that run stays below the peak, so the fitted and closed-form slopes agree there), while the  $D$  reaching that same product ranges over 0.69–2.00.  $D$  must therefore not be asked of this fit at all. It is measured on the *recorded* trace by the brush estimator of Section 5.5.2, which fits  $F_y(\alpha)$  directly and is not subject to Eq. (13), then held fixed while  $B$ ,  $C$  and  $E$  carry the stiffness the rollout does determine.

### 5.4.2 The loop is not a contraction

Stage 4 makes each iteration's answer the nominal model stage 2 rolls out, and stage 3 then fits the trajectory that model produced, over a steering range already extrapolating  $1.5\times$  past the observed one. The fit reads its own forces back, so the loop is a positive feedback path on its own extrapolation error rather than a contraction: replayed offline on five recorded buffers, axle force RMSE against the plant's telemetry degrades monotonically with iteration count, 38.9, 53.0, 58.5 and 70.4 N at the

front for  $N_{it} \in \{1, 2, 3, 6\}$ . Pinning  $D$  does not remove this; it relocates it into  $B$ ,  $C$  and  $E$ . The remedy is to damp the step, not to shorten the loop, so stage 4 becomes a relaxed fixed-point iteration

$$\Phi_p^{(i)} \leftarrow (1 - \beta) \Phi_p^{(i-1)} + \beta \tilde{\Phi}_p^{(i)}, \quad (14)$$

with  $\beta = 0.20$ ;  $\beta = 1$  recovers the inherited update exactly. The sweep over  $\beta$  (Section 6.8) carries the control the contribution needs:  $\beta = 0$  freezes the coefficients at the pinned prior, performing no identification at all, and is the worst setting. The iterations do carry information, but only a damped update preserves it.

Fitting itself applies Eq. (7) to the synthetic data under box constraints  $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$  per axle, over  $B$ ,  $C$  and  $E$  with  $D$  held. Relative to the baseline we raised the jittered multi-starts from 1 to 8 and added a derivative-free simplex [41] and a global differential-evolution solver [42] alongside trust-region reflective [43], through the same library [44]. Neither change addresses the degeneracy, and Section 6.2 verifies both as secondary.

## 5.5 Measurement-side corrections

### 5.5.1 Achieved steering, measured mass and geometry

$\alpha_f$  in Eq. (3) was built from the `/drive` setpoint. The measured achieved/commanded ratio is 0.76, so  $\alpha_f$  was systematically  $\approx 25\%$  too large and the fitted front stiffness  $\approx 20\%$  too small (13725 N/rad against 17198 N/rad fitted to the simulator's own per-wheel forces over the same run); using `feedback/steering_angle` recovers 18486 N/rad. Separately,  $m$  and  $l_{wb}$  come from the measured static wheel loads (Table 1) rather than a configuration file, and  $l_{wb} \equiv l_f + l_r$  is enforced by a regression test, because Eq. (7) uses the wheelbase for the normal loads and the axle split for the force division, and the two disagreeing is a silent bias.

### 5.5.2 Proprioceptive friction warm-start

The inherited pipeline had no prior for the peak factor:  $D$  started from a configuration constant and was left to the fit. Newton–Euler on the single-track body of Eqs. (1)–(2), solved for the axle forces rather than the accelerations, gives

$$F_{y,f} = \frac{I_z \dot{\omega} + l_r m a_y}{L \cos \delta}, \quad F_{y,r} = \frac{l_f m a_y - I_z \dot{\omega}}{L}, \quad (15)$$

which needs the IMU's lateral acceleration and the yaw acceleration and no tire model; vertical loads carry the longitudinal transfer  $F_{z,f/r} = m(g l_{r/f} \mp a_x h_{cg})/L$  [33], and yaw acceleration is the derivative of a local polynomial fit to the yaw rate (Savitzky–Golay, window 21, order 2) [45] rather than a difference, because  $I_z \dot{\omega}$  is a  $\approx 10\%$  term in  $F_{y,f}$  here.

Per axle, cornering stiffness and peak friction are then fitted jointly to the Fiala brush model [4, 24, 46]

$$F_y = \mu F_z (1 - (1 - \xi)^3) \operatorname{sgn} \alpha, \quad \xi = \min\left(\frac{C_\alpha |\tan \alpha|}{3\mu F_z}, 1\right), \quad (16)$$

by bounded nonlinear least squares with a soft- $\ell_1$  loss [43, 44], over samples with  $|a_x| \leq 0.35 g$ , the fit being pure-lateral while combined slip shrinks the lateral capacity [15]. The brush model replaces Eq. (4) here because it has two parameters, both of which the manoeuvre can support, against the four of the model whose non-identifiability at low slip this paper is about. What beats the utilisation bound is that  $\mu$  enters Eq. (16) through the *curvature* of  $F_y(\alpha)$  and not only through its asymptote: writing  $r = |F_y|/(C_\alpha |\tan \alpha|)$  for the force's departure from its own linear extrapolation, Eq. (16) inverts to

$$\mu = \frac{2k}{3 - \sqrt{3(4r - 1)}}, \quad k = \frac{C_\alpha |\tan \alpha|}{3F_z}, \quad (17)$$

so a 30 % departure from linearity ( $r = 0.7$ ) already pins  $\mu$ , well below the limit.

Eq. (17) also states its own failure mode:  $d\mu/dr$  grows without bound as the curve stays linear. The estimate is released only when four conditions hold: absolute grip demand  $\max |F_y|/F_z \geq 0.40$ , peak friction utilisation  $\max |F_y|/(\hat{\mu} F_z) \geq 0.40$ , relative precision  $\sigma_{\hat{\mu}}/\hat{\mu} \leq 0.15$  from the Gauss–Newton covariance, and  $\hat{\mu}$  not resting on its own search bound,  $[0.2, 2.0]$ ; otherwise the module logs which condition failed and falls back to the utilisation bound. The first two are the same quantity with and without the fit in the denominator, and both are needed: the utilisation test divides by the *fitted*  $\hat{\mu}$  of the same least-squares call, so an axle whose  $\mu$  comes out too low clears the gate on its own error, while the absolute ratio has no such freedom. The 40 % floor is the level the friction-estimation literature converges on [15].

### ***Applied two-sided, then pinned***

The released  $\hat{\mu}$  replaces  $D$  on both axles, in either direction, and  $D$  is then held fixed for the whole co-identification loop. Both axles take the *front* fit: the rear slip angle does not carry  $\delta$ , so on this trajectory the rear force curve has too little curvature to support Eq. (17) and its  $\mu$  comes out high (1.495/1.383/1.257 rear against 1.011/0.964/0.837 front on three admitted buffers of a  $\mu = 1.05$  run). This is not a yaw-inertia error: in quasi-steady cornering  $F_{y,f}/F_{y,r} \rightarrow l_r/l_f$ , the same ratio as  $F_{z,f}/F_{z,r}$ , so  $\mu_f = \mu_r$  whatever  $I_z$  is, and there is one road surface. It is two-sided rather than a floor, the rule it replaces having protected against a  $D$  that *moved* between cycles rather than one that was low; it is pinned and not merely seeded, since Eq. (13) leaves  $D$  free along  $BCD = \text{const}$  and on a live run a 0.785 seed ended at the 2.000 bound within one cycle; and an axle with no released  $\hat{\mu}$  holds the  $D$  it came in with, since there is nothing in the rollout to fit it against.

Pinning puts the whole weight of the result on the gate, and  $\sigma_\mu/\mu$  from the fit covariance is a *precision* figure, not an accuracy one: a wrong mass, yaw inertia, IMU bias or road bank all fit tightly and wrongly through Eq. (15). Here the configuration file's 240.0 kg against the 269.6 kg the wheels actually carry biased  $\mu$  11 % low until

the measured value replaced it, and the configured wheel coefficient is not the effective one the physics step uses, the server multiplying it by the road surface’s 0.70.

## 5.6 Deployment and validation protocol

The identifier runs as a single ROS 2 node cycling a rolling FIFO buffer through four states (collect, train, await the controller’s acknowledgement, run), re-identifying every  $T_{\text{reid}} = 30$  s. Coefficients are submitted over a service call whose non-acknowledgement leaves the previous model in place, so a rejected identification degrades to the last accepted model rather than to none. That acknowledgement also arms the scoring: a submission the controller *rejects* on its admissibility gate (Section 7) is a model the car never drove on, and the seconds between a fit completing and its acknowledgement landing are seconds still driven on the previous model.

Three levels of validation are used, in increasing strength, and all three score the *accepted* model only. One-step prediction on a held-out run, as in the baseline: RMSE and  $R^2$  of  $v_y$  and  $\omega$ , with online statistics in Welford’s form [47] and samples admitted only above 1 m/s. Direct force comparison against `feedback/tire.forces`: the identified curve is evaluated at the measured  $(\alpha, F_z)$  and compared per axle against the measured  $F_y$ , and against the plant’s own curve Eq. (8) at the same points, which separates the error of the fit from the error of representing a PhysX tire. Closed-loop tracking under the MPC: lateral deviation, its time-weighted integral, heading error and solver cost, split by active controller so the pure-pursuit acquisition phase is not scored against a model it does not use.

# 6 Experiments and Results

## 6.1 Setup

All experiments run on the architecture of Section 4: the Formula AI vehicle in CARLA driven through ROS 2 Humble, the server in synchronous mode at a 0.033 s fixed physics step. The identifier collects 30 s of data on a 50 Hz timer that the server’s step rate holds to 30 Hz (Section 4.4) and re-identifies from the newest buffer, matching the baseline’s data budget [6]. Ground truth is the simulator’s own per-wheel telemetry, which the identifier never reads; every figure is exported with the CSV holding its samples, and the cross-run comparison reads those CSVs rather than re-deriving any metric.

Two clocks appear below, and no metric mixes them. The controller node stamps its own quantities (run duration, handover time, lap time and every tracking error) on the ROS wall clock. The identifier and the force benchmark stamp theirs on the simulation clock, which the server advances faster than wall time, at a ratio between 1.28 and 1.66 across the four runs. Table 5 labels each block with the clock it is measured on.

### 6.1.1 Two friction schedules

Four runs are reported, two configurations crossed with two surfaces.

On the *static* surface the per-wheel coefficient is held at its configured 1.0, and the road-surface factor of 0.70 PhysX applies on top of it makes the effective coefficient 0.700, constant to the last digit the simulator reports over all force-validation samples of both runs. It is deliberately not the  $\mu = 1.05$  surface the tire prior was measured on (Section 6.1.4), so an arm that cannot move off its prior cannot score well. What it does not test is tracking: the identifier is asked to find one number and then hold it.

The *decaying* surface supplies the change. A schedule node drives the per-wheel coefficient over the ROS wall clock through

$$\mu_{\text{cmd}}(t) = \max(\mu_{\text{floor}}, \mu_0(1 - \eta t)), \quad \eta = 10^{-3} / \text{s}, \quad (18)$$

with  $\mu_0 = 1.0$  and  $\mu_{\text{floor}} = 0.5$ , reached at  $t = 500$  s; the command is applied through the bridge's `set_tire_friction` service while the car is driving, so the surface changes under a moving vehicle rather than between runs. The effective axle coefficient the plant reports therefore falls monotonically from 0.700 to the 0.350 floor, half the grip gone over the run, and the force-validation window opens at 0.637. The linear-decay protocol is that of LLA-MPC [9], which decays  $\mu$  at 2 % of nominal per second on a 1:43 vehicle. We keep the law and the total excursion and reduce the rate by a factor of 20, because the scaling of the platform sets what an on-track identifier can be scored against at all: at 2 % per second the surface is exhausted in 50 s, shorter than one 30 s buffer plus the 20.9 s its fit takes (Section 6.9), and no identifier of this class would publish a second coefficient set before the experiment ended. At Eq. (18)'s rate the surface instead loses about 0.06 of  $\mu$  per identification cycle, which is the regime in which tracking is a question rather than a race against the sampling period.

### 6.1.2 Reference trajectory

All four runs follow the same optimised raceline (Figure A1), generated with the TUM global race trajectory optimisation package [48] under its minimum-curvature formulation [32]: a closed 5828.2 m lap of 2916 waypoints, reference speed 9.62–25.78 m/s, minimum corner radius 18.7 m ( $|\kappa|_{\text{max}} = 0.0536 / \text{m}$ ). Its peak lateral demand is  $\max(v_x^2 |\kappa|) = 4.98 \text{ m/s}^2$  (0.51  $g$ ), roughly 73 % of the 0.70  $g$  the static surface sustains. On the decaying surface that same trajectory stops being admissible part way through the run, which is the point of running it: the plant passes below the reference's demand at an effective  $\mu$  of about 0.51, and what the two arms do about it is the result of Section 6.7.

### 6.1.3 The two configurations

The two configurations of the pipeline differ in the nine settings of Table 3 and in nothing else; vehicle, map, trajectory, controller, data budget, loop count, tire prior and box constraints are shared, and each configuration meets both surfaces with the same parameter file. **Baseline-NN-MSE** is the inherited configuration evaluated at full scale on the corrected rollout speed, so the comparison isolates those nine settings rather than reproducing the published pipeline in full; **Excitation-S4D-Pinn** is the shipped configuration. Since the settings move together, what follows is a configuration comparison, not a per-factor ablation, which Section 6.8 supplies offline for the four

that carry the result. Nor is it a test of this paper’s central contribution: the rollout speed is taken from the measured lap in *both* arms, and no run’s mean speed falls below 16 m/s, so **Excitation-S4D-Pinn**’s 15 m/s floor never binds. The excitation correction itself, together with the operating point it ships, is established in Section 6.2.

**Table 3** The two configurations. Each is run on both surfaces.

| Setting                   | Baseline-NN-MSE         | Excitation-S4D-Pinn             |
|---------------------------|-------------------------|---------------------------------|
| Residual architecture     | MLP ( <b>baseline</b> ) | S4D ( <b>s4</b> )               |
| Residual objective        | One-step MSE            | Physics-informed, Eq. (9)       |
| Model integrator          | Euler                   | RK4                             |
| Sweep amplitude           | Fixed 0.40 rad          | $1.5P_{99}( \delta )$ , clipped |
| Pacejka solver            | Trust-region [43]       | Differential evolution [42]     |
| Tikhonov weight $\lambda$ | 0                       | 0.05, frozen target             |
| Friction warm start       | Off                     | On; $D$ pinned per axle         |
| Fast- $\mu$ derate        | Off                     | On, 1 Hz on 6 s                 |
| Update relaxation $\beta$ | 1.00 (undamped)         | 0.20                            |

Everything not listed is identical between the two, the tire prior and the box constraints included.  $\beta$  is the relaxation of Eq. (14); 1.00 is the inherited undamped update. The fast- $\mu$  derate is the brush estimator of Section 5.5.2 refitted on a trailing 6 s window between coefficient cycles and published as an optional ratio the controller applies to the identified grip ceiling, clamped to [0.4, 1.2]; with the warm start off there is nothing to publish. The rollout speed is taken from the measured lap in both arms and is not a difference here.

#### 6.1.4 Provenance of the tire prior

The prior  $\Phi_p^0$  both arms start from, and that the Tikhonov term of Eq. (12) pulls toward, is the measured tire of Table 2, obtained by fitting Eq. (4) to CARLA’s own per-wheel telemetry over a dedicated limit-excitation run that reached  $\alpha = 0.37$  rad, axle  $\mu = 1.05$  and  $1.50 g$ . It is not independent of the validation data, and the manoeuvre that produced it is one the pipeline is not allowed to run, the raceline peaking at  $0.51 g$ . Decisively, it was measured on a surface none of these runs use, carrying  $D = 1.009/1.002$  and axle slopes of 13.2 and 13.9 kN/rad against the 0.70 and the 17.4 and 16.6 kN/rad the static surface realises. Returning it unchanged is a 44 % grip overstatement on the static surface and a 188 % one at the decaying surface’s floor, so an arm has to move away from that prior, in the right direction, to score well, and on the decaying surface it has to keep moving.

## 6.2 Selecting the rollout operating point ( $v, \delta_{\text{sweep}}$ )

This grid is where the rollout configuration the pipeline ships is *chosen*, not an auxiliary check on a choice made elsewhere: the 15 m/s speed floor and the  $\delta_{\text{max}} = 0.54$  rad clip of Section 5.2 are read off Table 4, and the reachable-friction bound Eq. (11) is what makes the search one-dimensional rather than a blind scan. The search is run where the answer is known exactly rather than in CARLA, since a CARLA lap cannot be scored against a  $D$  nobody knows. A tire with a known peak factor  $D$  is rolled out through the synthetic sweep of stage 2 (the shipped rollout, residual network set to zero) and fitted back with the shipped solver, *unregularised* and started from the true

coefficients, so everything it loses is lost to excitation. Rollout speed  $v$ , sweep amplitude  $\delta_{\text{sweep}}$  and true  $D$  are swept, each cell repeated over five noise realisations at the one-step persistence RMSE measured on CARLA runs of this vehicle. The configuration axes are swept one at a time, because  $\mu_{\text{reach}}$  is a function of both. The selection criterion is the one the pipeline needs: the widest range of true  $D$  recovered without railing, over the whole 0.70–1.80 column span, since the tire is unknown when the configuration is fixed. The span matters twice over on a decaying surface, where one configuration has to cover every  $D$  the run passes through.

**Table 4** The search that fixes the shipped rollout operating point: peak factor  $\hat{D}$  recovered from a known tire, swept along both arguments of  $\mu_{\text{reach}}$ . The shipped  $(v, \delta_{\text{sweep}})$  is selected from this grid.

| swept                                                                   | $\mu_{\text{reach}}$ | $\hat{D}$ (railed of 8) at true $D$ of |                  |                  |                  |
|-------------------------------------------------------------------------|----------------------|----------------------------------------|------------------|------------------|------------------|
| value                                                                   |                      | 0.70                                   | 1.05             | 1.40             | 1.80             |
| (a) rollout speed $v$ [m/s] swept at $\delta_{\text{sweep}} = 0.40$ rad |                      |                                        |                  |                  |                  |
| 4                                                                       | 0.43                 | 0.52±0.036 (1.8)                       | 0.57±0.013 (2.8) | 0.59±0.018 (3.8) | 0.50±0.001 (7)   |
| 6                                                                       | 0.96                 | 0.68±0.001 (1)                         | 1.04±0.014 (1.2) | 1.22±0.066 (1.8) | 1.26±0.097 (2)   |
| 8                                                                       | 1.70                 | 0.69±0.001 (1)                         | 1.02±0.001 (0.8) | 1.36±0.002 (0)   | 1.79±0.025 (1.2) |
| 13                                                                      | 4.49                 | 0.72±0.001 (0)                         | 1.05±0.002 (0)   | 1.38±0.001 (0.4) | 1.77±0.002 (0.4) |
| 20                                                                      | 10.64                | 0.92±0.004 (1)                         | 1.14±0.005 (0.4) | 1.43±0.002 (0)   | 1.80±0.005 (0)   |
| (b) $\delta_{\text{sweep}}$ [rad] swept at $v = 15$ m/s                 |                      |                                        |                  |                  |                  |
| 0.03                                                                    | 0.45                 | 0.80±0.019 (2)                         | 1.10±0.041 (2)   | 1.08±0.182 (2)   | 1.09±0.397 (2)   |
| 0.06                                                                    | 0.90                 | 0.92±0.003 (2)                         | 1.30±0.008 (2)   | 1.68±0.017 (2)   | 1.80±0.232 (3.4) |
| 0.15                                                                    | 2.24                 | 0.99±0.004 (2)                         | 1.38±0.003 (2)   | 1.71±0.019 (2)   | 1.98±0.009 (1)   |
| 0.40                                                                    | 5.98                 | 0.76±0.009 (1)                         | 1.05±0.002 (0)   | 1.39±0.002 (0)   | 1.78±0.001 (0)   |
| 0.54                                                                    | 8.08                 | 0.83±0.004 (1)                         | 1.08±0.002 (0)   | 1.38±0.001 (0)   | 1.78±0.008 (0)   |

Unregularised, started from the true coefficients. Each cell is  $\hat{D} \pm$  the half-width of its 95 %  $t$ -interval over five noise realisations, with the mean number of the eight coefficients resting on a box constraint in parentheses. The two axes of one rollout are not independent, so each realisation contributes the mean of its own  $(\hat{D}_f, \hat{D}_r)$  and  $n = 5$ .

$\mu_{\text{reach}}$  is Eq. (11) at  $L = 1.5334$  m; the inherited configuration is the first row of (a). It is the Ackermann *upper* bound and above  $\mu_{\text{reach}} \approx 2$  detaches from what the rollout realises, so a large value means “far past saturation”, not a demand the tire ever sees.

The grid says three things. First, the inherited 4 m/s rollout is uninformative about  $D$  at *every* grip level tested: the 0.70, 1.05, 1.40 and 1.80 tires all come back as  $\hat{D}$  between 0.50 and 0.59, with two to seven of the eight coefficients on a bound, so the fit reports  $\mu_{\text{reach}}$  and not the tire. Second, the boundary is  $\mu_{\text{reach}} \gtrsim D$  and not a speed: at 6 m/s, where  $\mu_{\text{reach}} = 0.96$ , the 0.70 and 1.05 tires are recovered to within 3 % while the 1.40 and 1.80 tires are 13 % and 30 % short, and for  $\mu_{\text{reach}}$  between 1.7 and 4.5 every tire is recovered to within 4 % with no systematic railing. Nor is the bound a quantity to maximise: at  $v = 20$  m/s,  $\mu_{\text{reach}} = 10.64$ , the rollout sits far past saturation for every tire on the grid and  $\hat{D}$  re-biases upward, the 0.70 tire returning 0.92. Third, the failure is not the solver’s: multi-start, a global solver or a better start point cannot recover a coefficient the data does not contain.

Panel (b) sweeps the other argument of Eq. (11). Starving the amplitude destroys  $D$  just as starving the speed does, so speed is not the governing variable, and the rollout rather than the tire sets the direction of the error: an under-speeded rollout returns *too little* grip, an under-swept one too much (at  $\delta_{\text{sweep}} = 0.06$  rad the 1.05 tire

comes back at 1.30 and the 1.40 tire at 1.68, two of the eight coefficients on a bound throughout those rows). That is the signature of an unidentified parameter, not a biased one. The axes do *not* collapse onto a common  $\mu_{\text{reach}}$ , for the upper-bound reason of Section 5.2: at  $v = 20$  m/s Eq. (11) reads 10.64 against a realised 1.40–2.18. Read against the realised demand both axes behave alike, so Eq. (11) is a cheap screening bound, conclusive when it falls below  $D$ , and not a measure of excitation achieved.

The two panels fix the shipped point between a lower edge and an upper one. On speed, railing loses the 1.40 and 1.80 tires at  $v \leq 6$  m/s and the saturation re-bias spoils the 0.70 tire at 20 m/s; only  $v = 8$  and  $v = 13$  recover every tire in the span to within 4%, and the 15 m/s floor sits inside that band with the measured lap speed governing above it. On amplitude,  $\delta_{\text{sweep}} \leq 0.15$  rad holds two coefficients on a bound at every  $D$ , while 0.40 rad and 0.54 rad rail nothing, so 0.54 rad is taken as  $\delta_{\text{max}}$ , the ceiling of the  $1.5P_{99}$  clip that selects the amplitude on a given lap. All of it is a best case: with the residual set to zero a Magic Formula is fitted to Magic-Formula data under exactly matched structure, so the selected point is the best setting *on this grid* and not a proven optimum.

891

### 892 6.3 Cross-run summary

893

Table 5 is the complete comparison over both surfaces. On the static surface **Excitation-S4D-Pinn** is better on every quantity scored against the plant and behind only on QP solve time, by 5% of the RMS and 1% of the worst case, both two orders of magnitude inside the control period. The margins are  $9.0\text{--}12.1\times$  on peak friction and axle force and  $1.5\times$  on MPC tracking error.

On the decaying surface the comparison stops being a margin and becomes an outcome. **Excitation-S4D-Pinn** completes the same three timed laps, at 0.139 m RMS lateral error under MPC, which is its static-surface figure to within 8% although half the grip is gone by the end. **Baseline-NN-MSE** leaves the track in lap 2 and the run is scored as a failure, its 23.7 m RMS error under MPC being the departure itself rather than a tracking quality. The peak-friction column is where the two diverge first: **Excitation-S4D-Pinn** follows the surface down, **Baseline-NN-MSE** publishes  $D = 2.000$  throughout, and the 1.408 error that implies is what the controller is holding when the trajectory stops being admissible. The rest of this section reads those rows against the figures; Section 6.8 attributes them.

909

### 910 6.4 Identified tire model

911

The coefficient block of Table 5 separates the two arms before any downstream metric does.

913

First, no coefficient of **Excitation-S4D-Pinn** rests on a box constraint in any accepted identification of either run. **Baseline-NN-MSE** puts  $D$  on its upper bound of 2.000 on both axes in *every* accepted identification of both runs and  $E_f$  on its lower bound of -3.0 likewise, and on the decaying surface it adds  $B$  on its lower bound of 4.0 on both axes, so five of its eight coefficients are resting on an edge in the set the car was driving on when it left the track. The *inner* fits are a separate matter: **Excitation-S4D-Pinn**'s static run logs 35 fits that stopped on a bound, 30 of them

920

**Table 5** Cross-run comparison. Two configurations on two surfaces.

| Metric                                                                   | Static surface, $\mu = 0.700$ |                       | Decaying surface, $\mu : 0.700 \rightarrow 0.350$ |                             |
|--------------------------------------------------------------------------|-------------------------------|-----------------------|---------------------------------------------------|-----------------------------|
|                                                                          | Excitation-S4D-Pinn           | Baseline-NN-MSE       | Excitation-S4D-Pinn                               | Baseline-NN-MSE             |
| <i>Outcome and closed-loop tracking (wall clock)</i>                     |                               |                       |                                                   |                             |
| Timed laps completed                                                     | <b>3</b>                      | <b>3</b>              | <b>3</b>                                          | <b>1</b> (off track, lap 2) |
| Run duration [s]                                                         | 580.4                         | 554.6                 | 640.9                                             | 358.7                       |
| Lap times [s]                                                            | 175.2 / 145.4 / 142.8         | 162.3 / 135.8 / 138.0 | 189.8 / 160.9 / 169.2                             | 167.7 / — / —               |
| RMS $e_y$ , overall [m]                                                  | <b>0.256</b>                  | 0.319                 | <b>0.264</b>                                      | 21.505                      |
| RMS $e_y$ , MPC [m]                                                      | <b>0.151</b>                  | 0.224                 | <b>0.139</b>                                      | 23.705                      |
| Max $ e_y $ , MPC [m]                                                    | <b>0.683</b>                  | 0.776                 | <b>0.664</b>                                      | 38.386                      |
| ITAE $e_y$ , MPC [m s <sup>2</sup> ]                                     | <b>15925</b>                  | 25384                 | <b>17888</b>                                      | $1.32 \times 10^6$          |
| MPC solve, RMS / max [ms]                                                | 0.399 / 2.604                 | <b>0.380 / 2.568</b>  | <b>0.420 / 3.589</b>                              | 0.678 / 4.062               |
| <i>Axle lateral force against per-wheel telemetry (simulation clock)</i> |                               |                       |                                                   |                             |
| Front / rear RMSE [N]                                                    | <b>27.2 / 33.2</b>            | 329.6 / 348.4         | <b>32.9 / 36.5</b>                                | 171.4 / 178.3               |
| Front / rear $R^2$                                                       | <b>0.993 / 0.987</b>          | -0.160 / -0.461       | <b>0.984 / 0.977</b>                              | 0.708 / 0.647               |
| Front / rear max abs. error [N]                                          | <b>264 / 135</b>              | 1295 / 1316           | <b>249 / 236</b>                                  | 2507 / 2069                 |
| <i>Peak fraction against the plant's effective <math>\mu</math></i>      |                               |                       |                                                   |                             |
| Front / rear $\mu$ RMSE, whole run                                       | <b>0.145 / 0.142</b>          | 1.300 / 1.300         | <b>0.209 / 0.207</b>                              | 1.408 / 1.408               |
| Front / rear $\mu$ RMSE, after release                                   | <b>0.078 / 0.078</b>          | 1.300 / 1.300         | <b>0.149 / 0.149</b>                              | 1.408 / 1.408               |
| <i>One-step state prediction</i>                                         |                               |                       |                                                   |                             |
| $v_y / \omega$ RMSE [m/s, rad/s]                                         | <b>0.014 / 0.049</b>          | 0.156 / 0.104         | <b>0.011 / 0.037</b>                              | 0.055 / 0.105               |
| <i>Last accepted coefficient set, front / rear</i>                       |                               |                       |                                                   |                             |
| $B$                                                                      | 12.89 / 13.75                 | 7.64 / 11.19          | 16.10 / 16.42                                     | <b>4.00 / 4.00</b>          |
| $C$                                                                      | 1.240 / 1.248                 | 1.806 / 1.346         | 1.362 / 1.416                                     | 1.204 / 1.591               |
| $D$                                                                      | 0.768 / 0.768                 | <b>2.000 / 2.000</b>  | 0.460 / 0.460                                     | <b>2.000 / 2.000</b>        |
| $E$                                                                      | -2.52 / -1.84                 | -3.00 / -1.77         | -2.08 / -1.40                                     | -3.00 / -1.74               |
| Axle stiffness [kN/rad]                                                  | 16.9 / 16.8                   | 37.9 / 38.4           | 13.8 / 13.6                                       | 13.2 / 16.2                 |
| Grip ceiling Eq. (6) [g]                                                 | 0.77                          | 2.00                  | 0.46                                              | 2.00                        |
| Plant grip at that moment [g]                                            | 0.70                          | 0.70                  | 0.35                                              | 0.53                        |

Bold marks the better arm within a surface for the metric blocks, and a coefficient resting on a box constraint in the coefficient block. The box is  $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$  per axle. The **Baseline-NN-MSE** decaying-surface row is the last set it published before leaving the track, and its tracking figures are the departure, not a tracking quality. Lap times run on the ROS wall clock while the simulator advances its fixed step faster than real time, at a ratio that differs between runs, so they are not read as a vehicle-performance comparison across arms; within the **Excitation-S4D-Pinn** column the static-to-decaying difference is discussed in Section 6.7.

on  $C$ 's lower 1.2 and none on  $D$ , which the pin takes out of the fit, against 96 for **Baseline-NN-MSE** spread over  $B$ ,  $C$  and  $D$ ; on the decaying surface the count rises to 56, still concentrated on  $C$ . It is the relaxation of Eq. (14) that keeps the accepted set off an edge its iterates touch. A peak factor of 2.000 against a plant realising 0.700 is less a 186 % parameter error than the box-edge signature of Section 3.3: the coefficient is not identified, and no per-coefficient bounds check distinguishes that from a fit.

Second, the linear range is identified rather than inherited. Every stiffness in the table is Eq. (5) at the static axle loads  $F_{z,f} = 1373 \text{ N}$ ,  $F_{z,r} = 1274 \text{ N}$ ; for the plant it is the slope fitted to the simulator's own per-wheel forces over  $|\alpha| \in [0.005, 0.02] \text{ rad}$ , so the two are comparable. The static surface realises 17.4 kN/rad front and 16.6 kN/rad rear over this run's load range, below the  $\bar{F}_z = 1$  closed-form values of Table 2 for the reason its first footnote gives. The prior carries 13.2 and 13.9, 24 % and 16 % low; **Excitation-S4D-Pinn** returns 16.9 and 16.8, 3.2 % low front and 1.4 % high rear, removing 87 % and 91 % of the prior's error, and the product  $BCDF_z$  that the rollout fit determines (Section 5.4.1) then holds within 16.0–16.9 kN/rad over the five cycles after the gate releases. **Baseline-NN-MSE** returns 37.9 and 38.4, 118 % and 132 % high, which is  $D = 2.0$  multiplying a  $B$  the fit chose against it. On the decaying surface **Excitation-S4D-Pinn**'s stiffness falls with the surface, to 13.8 and 13.6 kN/rad at the last accepted set, while **Baseline-NN-MSE** arrives at 13.2 and 16.2 kN/rad by a different route:  $B$  at its floor multiplied by  $D$  at its ceiling. A plausible stiffness assembled from two railed coefficients is the clearest case this paper has for gating on derived quantities rather than on coefficients, since the stiffness gate passes it.

Third, both sets keep the understeer sign and pass the stability gate of Section 7.1; what separates them is the grip ceiling, which Eq. (6) puts at 2.00  $g$  for **Baseline-NN-MSE** on both surfaces against 0.77  $g$  and 0.46  $g$  for **Excitation-S4D-Pinn**. The baseline's ceiling is a claim the plant cannot honour by a factor of 2.8 on the static surface and 3.8 at the moment the car leaves the decaying one, and the grip gate is one-sided, so it passes both. All of these readings are possible because the lap traverses part of the curvature:  $P_{99}(|\alpha|) = 2.7^\circ$  and  $\max |\alpha| = 5.8^\circ$  against the  $8.0^\circ$  at which the plant's curve peaks on the static surface, and  $\max |F_y|/F_z$  reaches 0.71 front and 0.72 rear in transients against 0.49 and 0.50 at  $P_{99}$ . This is not a limit manoeuvre, but it suffices for the friction gate to release, which on the  $\mu = 1.05$  runs it did not; as the decaying surface falls the same lap becomes relatively more demanding, so the gate keeps releasing rather than closing.

1001

## 1002 6.5 Peak friction: holding one surface, tracking another

1003

Figures 2 and 3 show what each arm published for the peak factor over the two runs, and Table 5 gives the errors behind it. The pinned- $D$  design of Section 5.5.2 is directly observable here, because both arms start 44 % away from the static answer and 188 % away from the decaying one's floor.

On the static surface **Excitation-S4D-Pinn** publishes six accepted identifications. The first carries  $D_f = 1.009$ ,  $D_r = 1.002$ : the friction gate rejected that buffer, so the axles held the prior they came in with, the hold-rather-than-fit rule of Section 5.5.2 and not a fitted value. From  $t = 140.6 \text{ s}$  the gate releases on every subsequent cycle and the published peak factor drops to 0.779, 0.821, 0.749, 0.756 and 0.768, front and

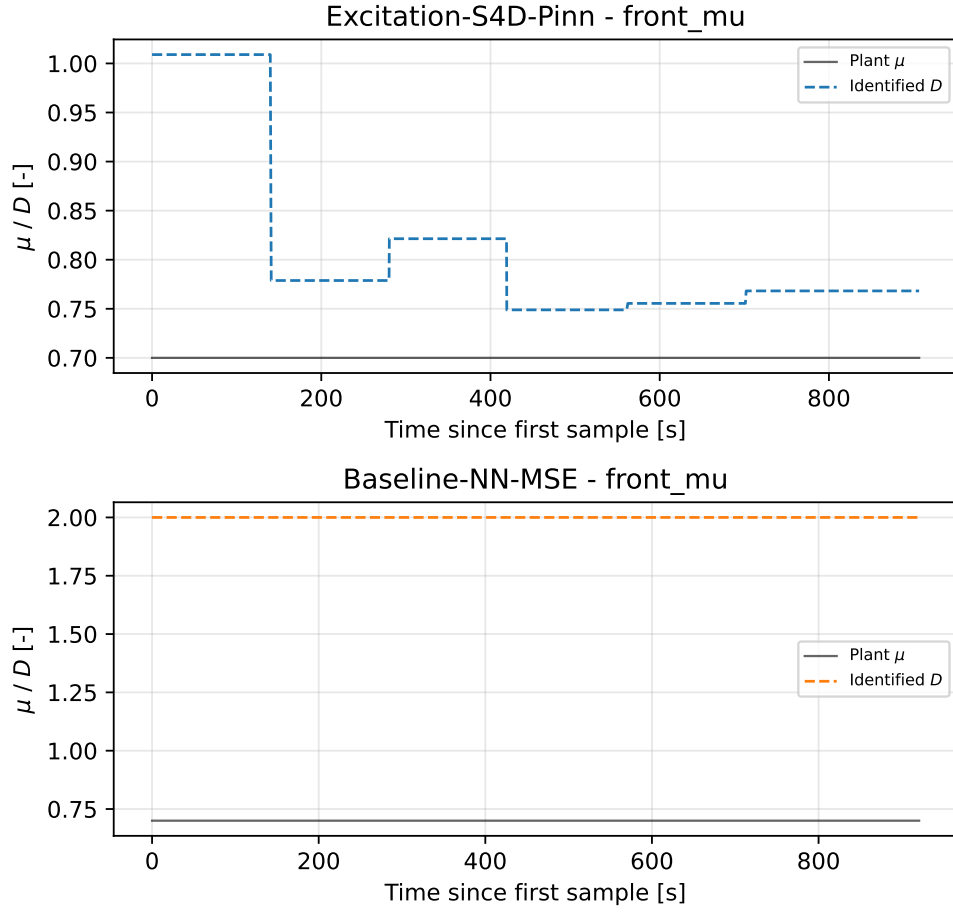
rear alike, against a plant at 0.700: 10.7 % high on average, from a prior that was 44 % high. The whole-run  $\mu$  RMSE is dominated by the first 140 s at the prior; after the first release it is 0.078 on both axles.

On the decaying surface the same six cycles land on a surface that has moved between each of them. The gate again rejects the first buffer and releases from  $t = 145.1$  s, and the published peak factor then falls monotonically: 0.706, 0.679, 0.609, 0.525 and 0.460, against a plant falling from 0.580 to 0.350 over the same interval. Read as a rate, the identifier recovers the decay itself: the published sequence falls at  $4.28 \times 10^{-4}$  /s against the plant's  $4.00 \times 10^{-4}$  /s, a 7 % error on the quantity that a static surface cannot test at all. Read as a level it carries two offsets, and separating them matters. Each cycle is fitted on the buffer that preceded it, so the published value describes the surface of roughly one cycle earlier; against the plant one cycle back the residual is 10.9–17.4 % high, mean 14.3 %, against the 10.7 % the same estimator carries on a surface that does not move. The apparent whole-run RMSE of 0.209 is therefore that familiar brush-model bias plus one cycle of transport lag on a surface losing about 0.06 of  $\mu$  per cycle, and not a new error mode. The remedy for the lag is a shorter cycle rather than a better fit, and the fast- $\mu$  derate of Table 3 is that shorter cycle; on this run it spent the last cycles at the upper end of its clamp and so did not shorten anything (Section 6.8).

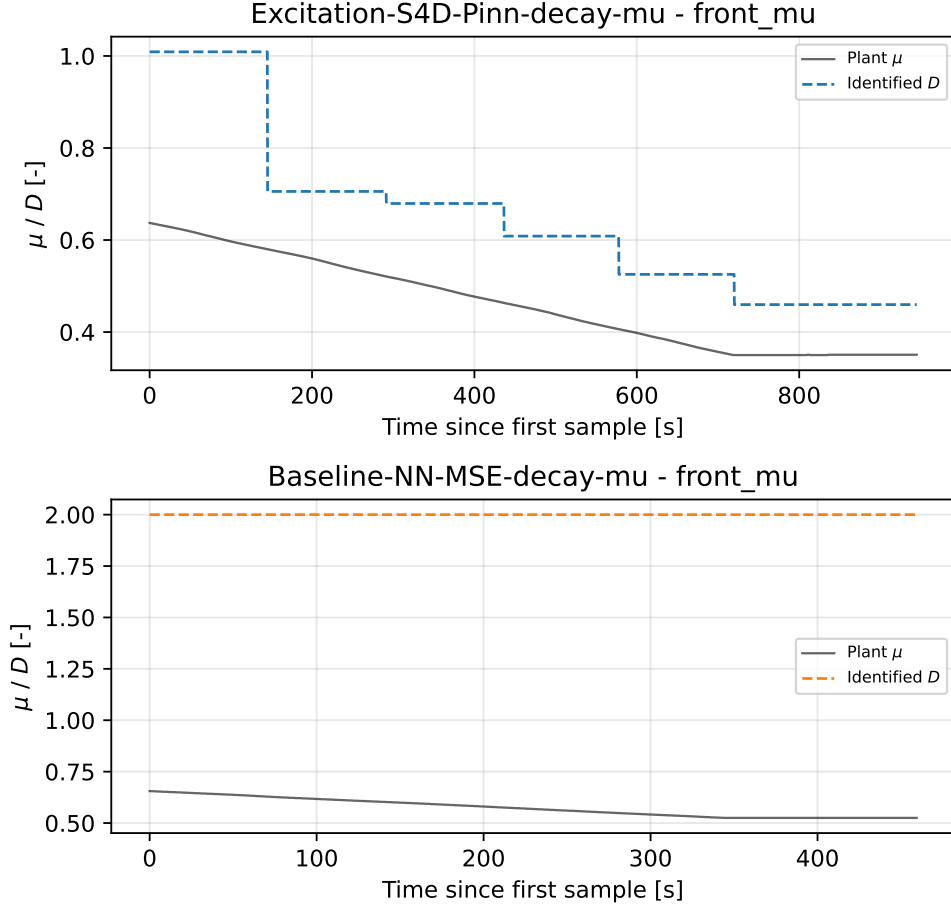
**Baseline-NN-MSE** returns  $D = 2.000$  on both axles in every accepted identification of both runs and never leaves the bound. On the static surface that is a constant  $\mu$  error of 1.300, so **Excitation-S4D-Pinn** is  $9.0\times$  better on the whole-run figure and  $16.7\times$  better once its warm start has released. On the decaying surface the error grows as the plant walks away from the bound, reaching 1.408 RMSE before the run ends early, and **Excitation-S4D-Pinn** is  $6.7\times$  better there. Two qualifications belong on this whole subsection: the peak factor is *measured* by the brush estimator on the recorded trace and then held, not recovered by the Pacejka fit, which Section 5.4.1 shows cannot recover it; and the residual bias is one-signed, consistent with the brush model's tendency to over-read  $\mu$  where the curve is not fully traversed, which is also why the tracking result below has margin to work with. Appendix B evaluates the same estimator on a synthetic brush-tire plant, where  $\mu$  is known exactly and the excitation can be swept independently of it, and shows that it is flat in excitation above its gate while the utilisation bound it replaces is not.

## 6.6 Axle lateral force and one-step state prediction

The identified curve is scored at the measured  $(\alpha, F_z)$  against the plant's own lateral force, summed per axle (Table 5). On the static surface **Excitation-S4D-Pinn** leads on both axles by an order of magnitude in RMSE and in the error tails; the baseline's negative  $R^2$  means its curve predicts the plant's axle force worse than the mean of that force does. This is the linear-range result seen as force: at  $P_{99}(|\alpha|) = 2.7^\circ$  the metric is dominated by cornering stiffness, so the models separate almost entirely on slope. Because  $D$  multiplies that slope in Eq. (5), **Baseline-NN-MSE's** railed  $D$  is the direct cause of its force error. Pinning  $D$  alone does not produce these numbers: with  $\beta = 1$  the same arm degrades monotonically with iteration count (Section 6.8), which is why the relaxed update of Eq. (14) belongs in the method rather than a configuration file.



**Fig. 2** Peak friction over the run on the static surface, front axle, one row per configuration: the plant's effective  $\mu$  (black) against the identified peak factor  $D$  published at each accepted identification. The rear axle is identical and is omitted, since  $D$  is pinned to the same value on both axles (Section 5.5.2). Every departure from the black line is estimator error: **Excitation-S4D-Pinn** publishes the  $\mu = 1.05$  prior while the gate is closed, steps down onto the plant within 11% at the first release, and holds. **Baseline-NN-MSE** sits on the  $D \leq 2.0$  box constraint throughout.



**Fig. 3** Peak friction over the run on the decaying surface, drawn exactly as Figure 2. The five accepted **Excitation-S4D-Pinn** cycles walk the published  $D$  down at 7% of the plant's own rate, offset by one cycle of transport lag and by the same estimator bias the static surface shows. **Baseline-NN-MSE** again sits on the  $D \leq 2.0$  box constraint, and its trace ends where the car left the track.

On the decaying surface **Excitation-S4D-Pinn**'s force error grows by about a fifth, 32.9N front against 27.2N, which is the transport lag of Section 6.5 converted into force, and  $R^2$  stays at 0.984 and 0.977. **Baseline-NN-MSE**'s force error is *lower* there than on the static surface, 171.4N against 329.6N, and its  $R^2$  turns positive. That is not a success: its run is scored over 2597 samples against 10744, it ends before the surface reaches its floor, and its worst-case error is the largest in the table at 2507N. A model with five coefficients on a box edge produced a better-looking aggregate over a shorter and slower run, which is the diagnostic problem Section 3.3 describes.

Level 1 of the validation protocol is reported against a persistence baseline (next state equals last measured state) written by the same benchmark node. **Excitation-**

1151 **S4D-Pinn** leads on both signals on both surfaces, by an order of magnitude on lateral  
1152 velocity on the static surface ( $R^2$  0.998 against 0.709) and by a factor of two on  
1153 yaw rate ( $R^2$  0.892 against 0.527). Neither model beats persistence at the 0.035 s  
1154 model step for either configuration or signal, because a one-step metric on a smooth  
1155 trajectory rewards not moving. One-step accuracy is therefore weak evidence for either  
1156 tire model, and the gap is read only as confirming the direction the axle-force and  
1157 closed-loop comparisons establish.

1158

## 1159 6.7 Closed-loop tracking

1160

1161 Figures 4 and 5 cover the whole of each run.

1162 On the static surface both arms finish three timed laps with zero emergency halts  
1163 and the same finite-state sequence, with the handover at  $t = 79.1$  s for **Excitation-**  
1164 **S4D-Pinn** and  $t = 64.0$  s for **Baseline-NN-MSE**. Under MPC, where the identified  
1165 model enters the loop, **Excitation-S4D-Pinn** tracks at 0.151 m RMS lateral error  
1166 against 0.224 m, a 32 % reduction, with maximum 0.683 m against 0.776 m. ITAE is  
1167  $\int t|e_y|dt$ , neither scale-free nor comparable across runs of different length, so the  
1168 duration-normalised forms  $IAE/T = 0.104$  against 0.157 and  $ITAE/(T^2/2) = 0.095$   
1169 against 0.166 are reported beside it; all three order the runs alike. The split by con-  
1170 troller confirms that the whole difference sits in the MPC phase: the pure-pursuit  
1171 phases, which do not use the identified model, give 110 and 107, within 3 % of each  
1172 other. The QP cost is unchanged, both arms staying two orders of magnitude below  
1173 the 50 ms control period; the identified coefficients change the model the QP carries,  
1174 not its size.

1175 On the decaying surface the two arms separate at  $t \approx 246$  s, in lap 2. **Baseline-**  
1176 **NN-MSE**'s lateral error steps from under 1 m to 38.4 m inside a few seconds, its speed  
1177 goes through zero to  $-16.9$  m/s, and the run ends on a collision whose horizontal  
1178 impulse of 415.9 N s is twice the 200 N s abort threshold. The plant's effective  $\mu$  at that  
1179 moment is 0.525, worth 0.53  $g$ , against a reference that is still asking for 0.51  $g$  at its  
1180 peak and a model that is still reporting a 2.00  $g$  ceiling. The failure is not a tracking  
1181 degradation that grew until it mattered; the tracking error is flat and small right up  
1182 to the departure, which is what a grip limit being crossed looks like from inside the  
1183 controller.

1184 **Excitation-S4D-Pinn** completes its three laps on the same surface and its tracking  
1185 error does not degrade as the grip halves: binned in 100 s windows after the handover  
1186 it reads 0.153, 0.138, 0.155, 0.094, 0.152 and 0.092 m RMS, with no trend, against  
1187 0.123 to 0.179 m over the corresponding windows of the static-surface run. What moves  
1188 instead is the speed. The identified grip ceiling falls with the published  $D$ , the controller  
1189 re-clamps the reference speed profile against it, and the peak lateral demand of the  
1190 clamped reference drops from 4.98 m/s<sup>2</sup> (0.51  $g$ ) at the start of the run to 2.92 m/s<sup>2</sup>  
1191 (0.30  $g$ ) at the end. The three lap times lengthen accordingly, to 189.8, 160.9 and  
1192 169.2 s against 175.2, 145.4 and 142.8 s on the static surface, an 18 % loss on the last  
1193 lap. That trade is the whole argument for identifying on track: the vehicle pays lap  
1194 time it has to pay, and the identification is what tells it how much. **Baseline-NN-MSE**  
1195 never makes the trade, because a peak factor pinned to a box constraint cannot tell the  
1196

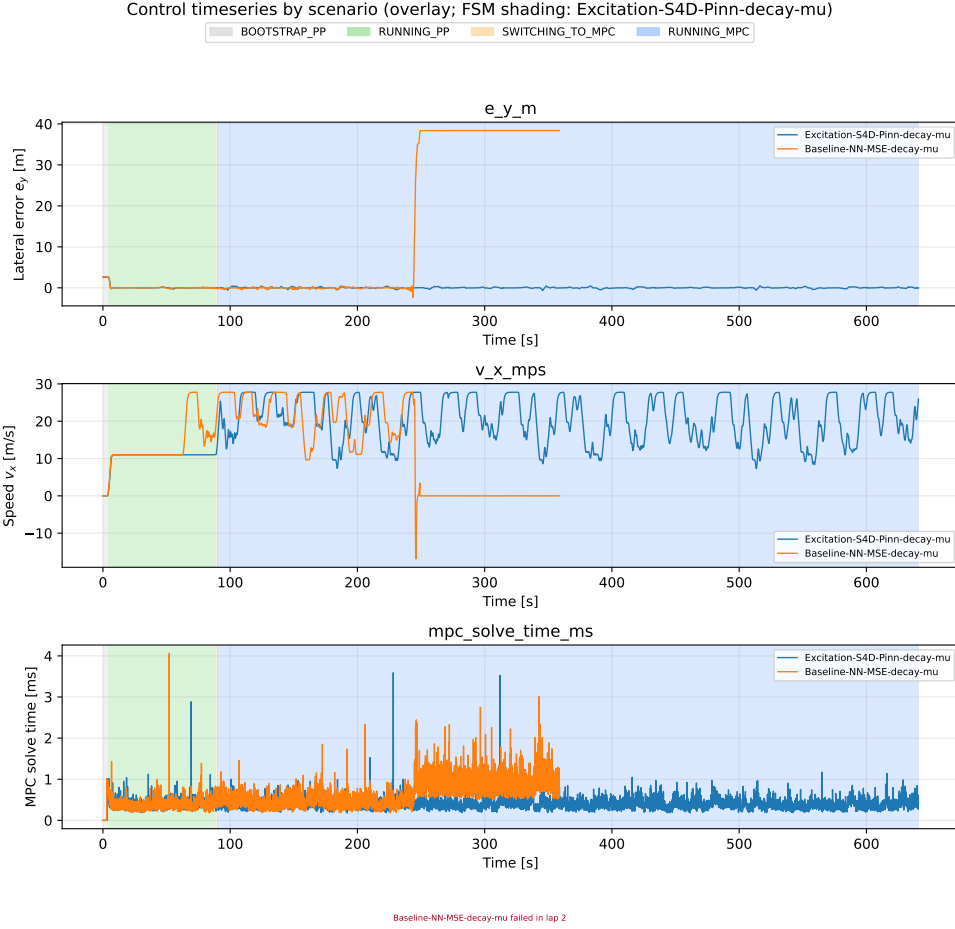
controller that anything has changed. Lap times themselves are not read as a vehicle-performance result across arms, for the clock reason of Section 6.1; the comparison that carries this paragraph is within the **Excitation-S4D-Pinn** column, where the same configuration meets two surfaces.



**Fig. 4** Lateral error, longitudinal speed and MPC solve time over the whole of the static-surface run, both configurations overlaid, shaded by the active controller state of the **Excitation-S4D-Pinn** arm. Both arms run to three laps and the traces stay superimposed. Solve time stays two orders of magnitude below the 50 ms control period.

## 6.8 What carries the result

The two arms differ in nine settings at once, so the comparison above establishes an ordering and attributes nothing. Appendix C attributes it by offline replay of the



**Fig. 5** The same three signals over the decaying-surface run, drawn exactly as Figure 4. **Baseline-NN-MSE** departs the track at  $t \approx 246$  s ( $e_y$  to 38.4 m, speed through zero) while **Excitation-S4D-Pinn** runs to 640.9 s. **Excitation-S4D-Pinn**'s speed trace loses its peaks over the second half of the run: the controller re-clamps the reference against a grip ceiling that is falling because the identifier says it is.

identification stage on the five 30 s buffers each arm collected during its own run, scored with the quantity the closed-loop tire-force benchmark reports. Four settings carry the result. Refusing to fit an unmeasured  $D$  removes most of the error, cutting the mean  $|D - \mu|$  from 0.585 to 0.084 on replayed buffers of a  $\mu = 1.05$  run. Damping the loop carries the rest: axle force RMSE is 70.4 and 62.3 N at the inherited  $\beta = 1.00$ , 92.5 and 71.8 N at  $\beta = 0$ , which is no identification at all, and 28.0 and 19.4 N at the  $\beta = 0.20$  that ships. Normalising the physics-informed objective is necessary but not sufficient alone, and the S4D residual is worth 21 % on the force metric at matched settings. Two settings buy nothing measurable: the residual *objective*, indistinguishable from a plain one-step MSE once the normalisation and the damping are both in place, and

the fast- $\mu$  derate, which sat at the upper end of its  $[0.4, 1.2]$  clamp over the last cycles of the decaying run, so the reference re-clamp of Section 6.7 is carried by the published  $D$  and not by the derate.

## 6.9 Identification cost

One S4D identification, six co-identification iterations on a 30 s lap, takes 20.9 s end to end, inside the 30 s re-identification period. The reduction from the 181.4 s it originally took is exact: no epoch count, iteration count, early-stopping rule or loss weight changed, and the identified coefficients agree to four decimal places. That margin is what makes the decaying-surface result possible at all, and it is also what bounds it: one cycle of lag on a surface moving at Eq. (18)’s rate costs the 10.9–17.4 % offset of Section 6.5, and a surface moving ten times faster would cost ten times that. The per-phase breakdown is in the supplementary material.

## 7 Integration with the Controller

The identified model leaves the pipeline through one interface. A nonlinear MPC path-tracking controller (acados [30], partial-condensing HPIPM [31], horizon  $N = 20$ , run at 20 Hz) carries the same single-track model as the identifier and consumes  $\Phi_p$  directly through Eq. (4), pushed over a ROS 2 service at the end of each identification cycle. We claim no contribution to the controller design; it appears here as the consumer that defines what an admissible identification is.

### 7.1 Admissibility gates

A coefficient set can lie strictly inside every box constraint and still be unusable. We observed two ways, each requiring a gate on a *derived* quantity rather than a coefficient:

1. **Grip ceiling versus trajectory demand.** The controller computes  $\max_k(v_{x,k}^2|\kappa_k|)$  over the speed-clamped reference, rejects any identification whose  $a_y^{\max}$  from Eq. (6) falls below it, and otherwise re-clamps the reference speed profile against that ceiling, which is the path by which a falling peak factor slows the vehicle in Section 6.7. On the earlier trajectory the degenerate fit reported a 0.40  $g$  ceiling against a 1.22  $g$  demand and produced 41.89 m of cross-track error at 27 m/s, against 0.03 m for a physically plausible set ( $D = 1.5$ ) on the same trajectory, at the same speed and through the same controller; the same identified set gives 0.16 m at 11.1 m/s, the  $v^2$  scaling the Introduction describes. These figures come from an offline closed-loop harness rather than a controlled pair of simulator runs, and are quoted as the observation that motivated the gate, not as a result. The gate is one-sided by construction and catches only a ceiling that is too low, so **Baseline-NN-MSE**’s 2.00  $g$  (Section 6.4) passes it on both surfaces, which is why the decaying-surface departure is visible in the controller and not at this interface.
2. **Understeer sign and critical speed.** A set with  $l_f C_f > l_r C_r$  whose  $v_{\text{crit}}$  falls inside the controller’s speed range makes the prediction model open-loop unstable, and the resulting  $\rho(A_d) > 1$  is raised to  $\rho^N$  in the condensed QP. An observed set gave  $C_f = 37031 \text{ N/rad}$  against  $C_r = 11691 \text{ N/rad}$ , i.e.  $v_{\text{crit}} = 14.5 \text{ m/s}$  at the

1335 measured geometry and mass of Table 1, inside the controller’s speed range on  
1336 every trajectory in this paper.

1337 A third observation is a controller-implementation detail rather than a gate on the  
1338 identification: an admissible set can still be stiff relative to the control period, so the  
1339 prediction model sizes its RK4 sub-step from its own fastest lateral mode, restoring  
1340  $\rho \approx 1.000$  across the speed range, at a median QP setup-and-solve cost of 0.36 ms  
1341 against the 50 ms control period.

1342 An identification that passes the gates is a model the controller can be held respon-  
1343 sible for; one that fails is reported as unidentified rather than deployed. Two further  
1344 controller-side changes were required before any identified model could be evaluated  
1345 at full-scale speeds: reference speeds are clamped at ingest, and the solved state is  
1346 rolled forward by the measured transport delay before each solve. Both are recorded  
1347 for completeness and are not identification results.

## 1349 8 Discussion and Limitations

### 1351 8.1 What generalises, and what does not

1352 The reachable-friction bound Eq. (11) is kinematic: it involves no tire parameter, no  
1353 simulator-specific quantity and no property of the residual model, so it applies to  
1354 any implementation of the virtual-steady-state-data strategy of [6], in simulation or  
1355 on hardware. Its practical content is a design rule: the rollout must be configured to  
1356 demand at least the friction one intends to identify, and where it cannot,  $D$  should  
1357 be reported as unidentified rather than fitted. The diagnostic signature, coefficients  
1358 resting exactly on box constraints, is likewise generic and should be checked and logged  
1359 by any bounded tire-fitting routine, being otherwise indistinguishable in the output  
1360 from a successful fit. The two structural properties of Section 5.4 generalise on the same  
1361 grounds: the  $BCD$  degeneracy follows from the quasi-steady force recovery Eq. (7),  
1362 and the absence of contraction from stage 4 alone. What does not generalise is the  
1363 value 0.20, a property of this vehicle’s extrapolation error; what does is that  $\beta = 1$  is  
1364 not defensible and  $\beta = 0$  worse still. The insufficiency of per-coefficient box constraints  
1365 is also platform-independent, and the decaying surface adds a third instance to the  
1366 two we had: a stiffness that looks right while being the product of a railed  $B$  and a  
1367 railed  $D$ , carried by a controller until the surface fell below the reference’s demand.

1368 Three results are properties of this plant and must not be read as statements about  
1369 tires: the numerical tire values, since the plant is a PhysX wheeled-vehicle model; the  
1370 gap between configured wheel friction and realised peak axle friction (1.0 against 0.70),  
1371 which we report because an identifier tuned to converge on the configuration file’s  
1372 number will converge on the wrong one; and the 0.76 achieved/commanded steering  
1373 ratio, where the lesson (identify against the achieved actuator state) is general but  
1374 the magnitude is not. Sagmeister *et al.* [27] show sensitivity to model fidelity is largest  
1375 near the acceleration limits, where this study operates, and R-CARLA [26] exists  
1376 because CARLA’s native dynamics are a known weak point for racing-grade work, so  
1377 hardware replication is necessary before any claim about identification accuracy on a  
1378 real vehicle.

1380

## 8.2 Limitations

1. **One run per configuration and surface, so no metric in Table 5 carries an uncertainty.** The 0.151 m against 0.224 m RMS tracking difference in particular has no confidence interval and no evidence that it exceeds run-to-run variation, which is not measured here; the order-of-magnitude force and friction margins are large enough that such variation is unlikely to reverse them, but that is an argument, not a measurement. The decaying-surface departure is a single event and identifies a failure *mechanism*, a grip ceiling pinned to a box constraint while the plant falls below the reference's demand, not a failure rate.
2. **One decay law, at one rate, on one vehicle and trajectory.** The schedule is linear at 0.1 % of nominal per second with a floor at half, a factor 20 slower than the 2 % per second of [9], for the reason Section 6.1.1 gives: a faster decay outruns the 30 s buffer this class of identifier is built on, so the experiment would measure the sampling period rather than the estimator. The runs therefore establish that the pipeline tracks a friction change slow relative to its own cycle, with the scaling of Section 6.9. Step changes, which the scenario file also carries, are not run, and the lag the decay exposes would appear on a step as an overshoot of the whole cycle.
3. **The two surfaces were run on different controller builds.** The decaying-surface pair carries a grip rate limiter, a binding-axle derate and the fast- $\mu$  hook of Table 3, none of which are in the static-surface pair. Within each pair the two arms are identical apart from Table 3, so each comparison is clean; across pairs, only the Excitation-S4D-Pinn lap-time and tracking-error trend is read.
4. **The friction estimator's excitation sweep is synthetic** (Appendix B), so the excitation dependence of its 11 % bias is not mapped against the simulator's per-wheel forces. The published peak factor is exactly this estimate, so the bias passes through undamped, and on the decaying surface it arrives with one cycle of lag on top of it. Relatedly, the rear axle is not identified on this trajectory and the front-axle  $\mu$  is applied to both (Section 5.5.2); the quasi-steady argument making  $\mu_f = \mu_r$  holds on a homogeneous surface and would not survive a split- $\mu$  one, which this sensor set cannot detect in the first place.
5. **The two configurations differ in nine settings at once** (Table 3), so Section 6.3 attributes nothing to any one of them. Section 6.8 does attribute four, but by offline replay on recorded buffers, so its ordering is not a closed-loop ordering, and the integrator, the sweep bound and the regularisation are not separated at all. The S4D figure there (21 % on force RMSE at matched settings) has the same status: a measured ablation, not a reproduction of the closed-loop S4 improvement of [21]. The excitation-aware rollout, this paper's central contribution, is active in *both* arms, so the closed-loop comparison is not evidence for the identifiability correction; the evidence for that is the sweep of Section 6.2.
6. **The physics-informed objective buys nothing measurable here.** Once its units are reconciled with the data term and the loop is damped, it is indistinguishable from a plain one-step MSE on this manoeuvre. It is retained because it is the paper's stated formulation and costs nothing, not because it is shown to help.
7. **Longitudinal dynamics are out of scope.** As in the baseline, only lateral dynamics are identified. Combined slip and load transfer are neglected, and the

simulator’s longitudinal wheel force channel is drivetrain torque rather than a contact-patch force, so it is unusable as ground truth for a longitudinal extension without further work.

1430

## 1431 9 Conclusion and Future Work

1432

This paper studied the identifiability of a learning-based on-track Pacejka identification pipeline [6] at full vehicle scale, using a CARLA simulation architecture that exposes the plant’s own per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, the pipeline fails in a specific, diagnosable and correctable way: a loss of excitation created by its own virtual-data step, which no change of solver, bounds or tuning removes, and whose reachable friction depends on the rollout configuration and not the tire. Two further properties of the loop follow from the same analysis. Stage 3 reconstructs its forces from the rollout’s own quasi-steady yaw balance, so its residual depends on  $B$ ,  $C$  and  $D$  only through their product and the peak factor is not recoverable from the rollout at any excitation; it must be measured on the recorded trace and held. And each fit is handed the trajectory its own previous answer generated, so stage 4 is a positive feedback path rather than a contraction and is reformulated as a relaxed fixed-point iteration at  $\beta = 0.20$ . The correction ships with an extrapolation bound on the steering sweep, a frozen regularisation target, identification against the achieved road-wheel angle, measured mass and geometry, and a dynamics residual normalised to the units of the data term, and it produces accepted identifications with no railed coefficient and the correct understeer sign.

A second surface settles what a single one cannot. With the plant’s friction driven down under the moving vehicle on the linear-decay protocol of [9] until half the grip is gone, the corrected pipeline recovers the decay rate to 7% and the level to 10.9–17.4% once one cycle of transport lag is accounted for; the falling peak factor re-clamps the reference from 0.51  $g$  of peak lateral demand to 0.30  $g$ , buying three completed laps at unchanged tracking error for 18% of lap time. The inherited configuration leaves the track in lap 2, holding a 2.00  $g$  grip ceiling against a plant offering 0.53  $g$ . Both outcomes have one cause: a peak factor resting on a box constraint cannot report that anything changed, so the identifier’s failure stays invisible until the controller acts on it.

Two qualifications stand on those numbers: the peak factor is identified by the brush estimator on the recorded trace, not by the Pacejka fit, which cannot identify it (Section 5.4.1); and each configuration is a single run per surface, so the closed-loop margins are observations rather than estimates with a confidence interval. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and place gates on derived physical quantities at that interface instead.

1467

### 1468 9.1 Future work

1469

1. **Shorten the cycle, or step the friction.** The decay runs expose one cycle of transport lag as the dominant error on a moving surface, worth 10.9–17.4% at the rate used here and proportionally more at a faster one. Two experiments follow

- directly, both scenario-file rows: the step schedules, on which the same lag appears as a bounded overshoot rather than a standing offset, and a shorter re-identification period, which the 20.9 s cycle cost of Section 6.9 leaves room for.
2. **Run the ablations of Section 6.8 in the loop.** The relaxation sweep, the  $D$ -pinning comparison and the S4D and objective ablations are offline replays, enough to attribute the result but not to confirm the ordering survives closed-loop coupling. Each is a scenario-file row.
  3. **Excite the peak, and design for it.** The lap reaches  $P_{99}(|\alpha|) = 2.7^\circ$  against a plant whose curve peaks at  $8.0^\circ$ , which is where the brush estimator’s 11 % bias comes from. Eq. (11) inverts into a requirement on the manoeuvre, so a controller that inserts bounded slalom or ramp-steer excitation when the identifier reports insufficient slip-angle coverage would move the pipeline toward active, safety-bounded experiment design [49].
  4. **Hardware replication.** The decisive test of every claim in Section 8 is a physical Formula Student vehicle. The identification code is unchanged between simulation and hardware; what changes is the availability of ground truth, which is what makes the simulation study a necessary precursor rather than a substitute.

## References

- [1] Liniger, A., Domahidi, A. & Morari, M. Optimization-based autonomous racing of 1:43 scale RC cars. *Optimal Control Applications and Methods* **36**, 628–647 (2015).
- [2] Kabzan, J., Hewing, L., Liniger, A. & Zeilinger, M. N. Learning-based model predictive control for autonomous racing. *IEEE Robotics and Automation Letters* **4**, 3363–3370 (2019).
- [3] Baumann, N. *et al.* ForzaETH race stack—scaled autonomous head-to-head racing on fully commercial off-the-shelf hardware. *Journal of Field Robotics* **42**, 1037–1079 (2024). Preprint: arXiv:2403.11784.
- [4] Pacejka, H. B. *Tire and Vehicle Dynamics* 3rd edn (Butterworth-Heinemann, Oxford, UK, 2012).
- [5] Kabzan, J. *et al.* AMZ driverless: The full autonomous racing system. *Journal of Field Robotics* **37**, 1267–1294 (2020).
- [6] Dikici, O. *et al.* Learning-based on-track system identification for scaled autonomous racing in under a minute. *IEEE Robotics and Automation Letters* **10** (2025). Preprint: arXiv:2411.17508. Code: <https://github.com/ForzaETH/On-Track-SysID>.
- [7] Ljung, L. *System Identification: Theory for the User* 2nd edn (Prentice Hall, Upper Saddle River, NJ, USA, 1999).
- [8] Dosovitskiy, A., Ros, G., Codevilla, F., López, A. & Koltun, V. *CARLA: An open urban driving simulator*. Proc. 1st Annual Conference on Robot Learning (CoRL), Vol. 78 of *Proceedings of Machine Learning Research*, 1–16 (PMLR, Mountain View, CA, USA, 2017). URL <https://proceedings.mlr.press/v78/dosovitskiy17a.html>.
- [9] AL-Sunni, M. F., Almubarak, H., Horng, K. & Dolan, J. M. LLA-MPC: Fast

adaptive control for autonomous racing (2025). URL <https://arxiv.org/abs/2505.19512>. 2505.19512.

[10] Bakker, E., Nyborg, L. & Pacejka, H. B. Tyre modelling for use in vehicle dynamics studies. SAE Technical Paper 870421, SAE International (1987).

[11] Wenzel, T. A., Burnham, K. J., Blundell, M. V. & Williams, R. A. Dual extended Kalman filter for vehicle state and parameter estimation. *Vehicle System Dynamics* **44**, 153–171 (2006).

[12] Sierra, C., Tseng, E., Jain, A. & Peng, H. Cornering stiffness estimation based on vehicle lateral dynamics. *Vehicle System Dynamics* **44**, 24–38 (2006).

[13] Rajamani, R., Piyabongkarn, D., Lew, J. Y., Yi, K. & Phanomchoeng, G. Algorithms for real-time estimation of individual wheel tire-road friction coefficients. *IEEE/ASME Transactions on Mechatronics* **17**, 1183–1195 (2012).

[14] Ahn, C., Peng, H. & Tseng, H. E. Robust estimation of road frictional coefficient. *IEEE Transactions on Control Systems Technology* **21**, 1–13 (2013).

[15] Acosta, M., Kanarachos, S. & Blundell, M. Road friction virtual sensing: A review of estimation techniques with emphasis on low excitation approaches. *Applied Sciences* **7**, 1230 (2017).

[16] Khaleghian, S., Emami, A. & Taheri, S. A technical survey on tire-road friction estimation. *Friction* **5**, 123–146 (2017).

[17] Hsu, Y.-H. J., Laws, S. M. & Gerdes, J. C. Estimation of tire slip angle and friction limits using steering torque. *IEEE Transactions on Control Systems Technology* **18**, 896–907 (2010).

[18] Hewing, L., Liniger, A. & Zeilinger, M. N. *Cautious NMPC with Gaussian process dynamics for autonomous miniature race cars*. Proc. European Control Conference (ECC), 1341–1348 (Limassol, Cyprus, 2018).

[19] Hewing, L., Kabzan, J. & Zeilinger, M. N. Cautious model predictive control using Gaussian process regression. *IEEE Transactions on Control Systems Technology* **28**, 2736–2743 (2020).

[20] Chrosniak, J., Ning, J. & Behl, M. Deep dynamics: Vehicle dynamics modeling with a physics-constrained neural network for autonomous racing. *IEEE Robotics and Automation Letters* **9**, 5292–5297 (2024). Preprint: arXiv:2312.04374.

[21] Wu, Z., Hu, C., Wang, Y., Xie, L. & Su, H. Vision-augmented on-track system identification for autonomous racing via attention-based priors and iterative neural correction (2026). URL <https://arxiv.org/abs/2603.09399>. 2603.09399.

[22] Gu, A., Goel, K., Gupta, A. & Ré, C. *On the parameterization and initialization of diagonal state space models*. Advances in Neural Information Processing Systems (NeurIPS) (New Orleans, LA, USA, 2022). Preprint: arXiv:2206.11893.

[23] Gu, A., Goel, K. & Ré, C. *Efficiently modeling long sequences with structured state spaces*. Proc. International Conference on Learning Representations (ICLR) (Virtual, 2022). Preprint: arXiv:2111.00396.

[24] Fiala, E. Seitenkräfte am rollenden luftreifen. *VDI-Zeitschrift* **96**, 973–979 (1954).

[25] NVIDIA Corporation. PhysX SDK 4.1 documentation: Vehicles — tire force computation (PxVehicleComputeTireForceDefault). Software documentation, PhysX SDK 4.1, as bundled with Unreal Engine 4.26 and linked by CARLA 0.9.16 (2019). <https://github.com/NVIDIA-Omniverse/PhysX/tree/release/104.2/physx>.

- [26] Brunner, M., Ghignone, E., Baumann, N. & Magno, M. R-CARLA: High-fidelity sensor simulations with interchangeable dynamics for autonomous racing (2025). URL <https://arxiv.org/abs/2506.09629>. 2506.09629.
- [27] Sagmeister, S., Kounatidis, P., Goblirsch, S. & Lienkamp, M. Analyzing the impact of simulation fidelity on the evaluation of autonomous driving motion control (2026). URL <https://arxiv.org/abs/2602.07984>. 2602.07984.
- [28] Abdo, J. *et al.* Simulating an autonomous system in CARLA using ROS 2 (2025). URL <https://arxiv.org/abs/2511.11310>. 2511.11310.
- [29] O’Kelly, M., Zheng, H., Karthik, D. & Mangharam, R. *F1TENTH: An open-source evaluation environment for continuous control and reinforcement learning*. Proc. NeurIPS 2019 Competition and Demonstration Track, Vol. 123 of *Proceedings of Machine Learning Research*, 77–89 (PMLR, Vancouver, BC, Canada, 2020). URL <https://proceedings.mlr.press/v123/o-kelly20a.html>.
- [30] Verschueren, R. *et al.* acados—a modular open-source framework for fast embedded optimal control. *Mathematical Programming Computation* (2022). Preprint: arXiv:1910.13753.
- [31] Frison, G. & Diehl, M. HPIPM: A high-performance quadratic programming framework for model predictive control. *IFAC-PapersOnLine (Proc. 21st IFAC World Congress)* **53**, 6563–6569 (2020).
- [32] Heilmeier, A. *et al.* Minimum curvature trajectory planning and control for an autonomous race car. *Vehicle System Dynamics* **58**, 1497–1527 (2020).
- [33] Rajamani, R. *Vehicle Dynamics and Control* 2nd edn (Springer, New York, NY, USA, 2012).
- [34] Bellman, R. & Åström, K. J. On structural identifiability. *Mathematical Biosciences* **7**, 329–339 (1970).
- [35] Raue, A. *et al.* Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics* **25**, 1923–1929 (2009).
- [36] Åström, K. J. & Wittenmark, B. *Adaptive Control* 2nd edn (Addison-Wesley, Reading, MA, USA, 1995).
- [37] Kaur, P., Taghavi, S., Tian, Z. & Shi, W. *A survey on simulators for testing self-driving cars*. Proc. 4th Int. Conf. on Connected and Autonomous Driving (MetroCAD), 62–70 (Detroit, MI, USA, 2021). URL <https://arxiv.org/abs/2101.05337>.
- [38] Kingma, D. P. & Ba, J. *Adam: A method for stochastic optimization*. International Conference on Learning Representations (ICLR) (San Diego, CA, USA, 2015).
- [39] Paszke, A., Gross, S., Massa, F. *et al.* *PyTorch: An imperative style, high-performance deep learning library*. Advances in Neural Information Processing Systems (NeurIPS) (Vancouver, BC, Canada, 2019).
- [40] Raissi, M., Perdikaris, P. & Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* **378**, 686–707 (2019).
- [41] Nelder, J. A. & Mead, R. A simplex method for function minimization. *The Computer Journal* **7**, 308–313 (1965).

- [42] Storn, R. & Price, K. Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization* **11**, 341–359 (1997).
- [43] Branch, M. A., Coleman, T. F. & Li, Y. A subspace, interior, and conjugate gradient method for large-scale bound-constrained minimization problems. *SIAM Journal on Scientific Computing* **21**, 1–23 (1999).
- [44] Virtanen, P., Gommers, R., Oliphant, T. E. *et al.* SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* **17**, 261–272 (2020).
- [45] Savitzky, A. & Golay, M. J. E. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry* **36**, 1627–1639 (1964).
- [46] Svendenius, J. & Gäfvert, M. A brush-model-based semi-empirical tire-model for combined slips. SAE Technical Paper 2004-01-1064, SAE International (2004).
- [47] Welford, B. P. Note on a method for calculating corrected sums of squares and products. *Technometrics* **4**, 419–420 (1962).
- [48] Institute of Automotive Technology, Technical University of Munich. `global_racetrajectory_optimization`. Software, Institute of Automotive Technology, Technical University of Munich (2019–2022). [https://github.com/TUMFTM/global\\_racetrajectory\\_optimization](https://github.com/TUMFTM/global_racetrajectory_optimization).
- [49] Goodwin, G. C. & Payne, R. L. *Dynamic System Identification: Experiment Design and Data Analysis* Vol. 136 of *Mathematics in Science and Engineering* (Academic Press, New York, NY, USA, 1977).
- [50] Foote, T. & Purvis, M. Standard units of measure and coordinate conventions. ROS Enhancement Proposal 103, Open Source Robotics Foundation (2010). <https://www.ros.org/reps/rep-0103.html>.
- [51] SAE International. Vehicle dynamics terminology. Surface Vehicle Recommended Practice J670, SAE International (2008).

## Appendix A Assumptions and Reproducibility Notes

The following are assumptions or design choices rather than measured facts, listed so a reader can judge how far each result depends on them.

*A1. Yaw inertia.*  $I_z = 51.1 \text{ kg m}^2$  is carried in the model file and is *not* measured. It enters Eq. (2) directly, so it scales the yaw-rate residual the network is trained on. It can be discharged by applying a known yaw moment and fitting  $I_z$  from the measured  $\dot{\omega}$ , or by reading the actor’s inertia tensor.

*A2. Centre-of-gravity split.*  $l_f = 0.738142 \text{ m}$ ,  $l_r = 0.795362 \text{ m}$  are taken from the controller configuration. The measured static load split (51.2/48.8) implies a marginally different split than the geometry does (51.9/48.1), a 1.4 % disagreement, below the level at which any conclusion here would change.

*A3–A4. CG height and steering sign.*  $h_{cg} = 0.3 \text{ m}$  is carried but not measured. The single-track identification model neglects load transfer, following the baseline, and its normal loads are static; the only consumer of  $h_{cg}$  is the longitudinal transfer term of the friction warm start (Section 5.5.2), where it moves the axle loads by up to 14 %

at the 0.35  $g$  the fit admits. `feedback/steering_angle` is published in degrees and treated as the front road-wheel angle, positive left, verified against the front-left wheel joint state to within 0.5 %; left and right road wheels are not distinguished.

*A5. Steady-state force recovery.* Eq. (7) is not a force measurement; being fully determined by  $v_x\omega$  it misattributes transients, and it is applied only to the *synthetic*, quasi-steady rollout. Its adequacy was checked on measured data against the simulator’s own axle force (correlation 0.993, slope 0.94).

*A6. The residual network is not the plant.* The corrected model is queried by the rollout at inputs the network has only partially seen. The extrapolation bound of Section 5.3 limits but does not eliminate this: the sweep still reaches  $1.5\times$  the observed 99th percentile of  $|\delta|$ , a choice that is empirical rather than derived.

*A7. Simulator determinism and timing.* CARLA runs in synchronous mode with a 0.033 s physics step; sim time advances faster than wall time, at a ratio set by the host load and therefore not identical across runs (between 1.28 and 1.66 on the four runs of Section 6). All identification and force-validation data is stamped from the simulation clock, while the controller node stamps its own metrics on the wall clock; the two are never mixed inside one metric (Section 6.1).

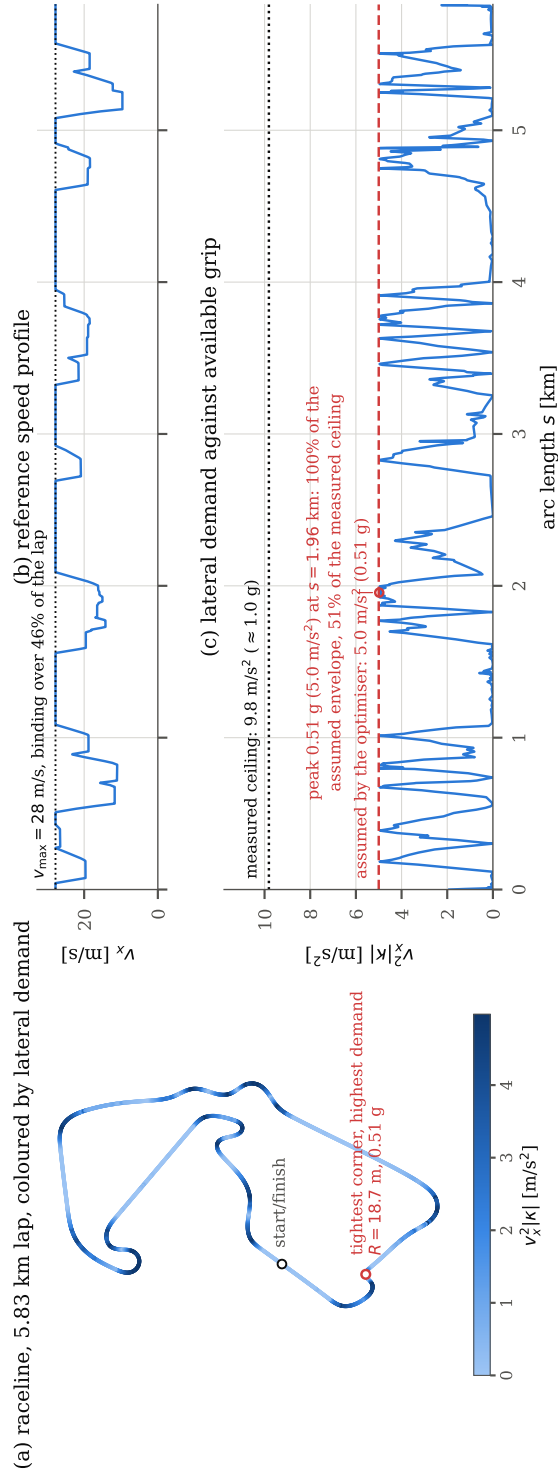
*A8. Software configuration of record.* ROS 2 Humble, Python 3.10, PyTorch with CUDA on an RTX 2080-class GPU; the bridge is a separate overlay; CARLA server 0.9.16 / Unreal Engine 4.26; acados 0.5.5 with partial-condensing HPIPM. The identifier’s acquisition timer runs at 50 Hz (the controller runs at 20 Hz, a 50 ms period) with a 30 s buffer, six co-identification iterations, 200 training epochs, full-batch Adam at  $5 \times 10^{-4}$ , and  $T_s = 0.035$  s. `nn.train` seeds PyTorch and NumPy at 42; `pacejka_solver.seed` is null in both shipped parameter sets, so the reported identifications are reproducible in distribution but not bit for bit, while the sweep of Table 4 is seeded explicitly and is bit-reproducible.

*A9. Reference trajectory.* Figure A1 shows the raceline all four runs follow and the lateral demand it places on the plant.

## Appendix B Peak-Friction Warm Start on a Synthetic Plant

The friction warm start is evaluated on a synthetic single-track plant with brush tires, because only a synthetic plant supplies a  $\mu$  known exactly and an excitation level that can be swept independently of it: a swept-sine steering input for 40 s at 50 Hz with true  $\mu = 0.70$ , the same effective peak as the static CARLA surface, the steering amplitude setting how far up the tire curve the manoeuvre reaches (Table B1).

At 63 % utilisation, a level an ordinary lap reaches, the utilisation bound is 37 % low while the brush fit is within 0.2 %, and it stays there across true  $\mu \in \{0.7, 1.05, 1.4\}$  on noise-free data; with odometry and IMU noise injected the error is grip-dependent, 20 % at 63 % utilisation and 0.4 % at the peak for  $\mu = 0.70$ . Below the absolute 0.40 floor on  $|F_y|/F_z$  the gate suppresses the fit and reports the bound, so the failure is visible rather than silent. The gate behaves the same way on the CARLA runs, rejecting the first buffer of each and returning  $\mu$  within 11 % of the static plant at an absolute axle utilisation of 0.54–0.68. That 11 % bias is the cost of a real rollout, a



**Fig. A1** The optimised raceline used for all four runs [32, 48]. (a) Track layout, coloured by lateral demand, with the marked corner ( $R = 18.7$  m) the tightest on the lap. (b) Reference speed profile. (c) Lateral demand  $v_x^2|\kappa|$  along the lap against the flat 5.0 m/s<sup>2</sup> envelope handed to the optimiser (dashed) and the grip the vehicle has on the static surface (dotted,  $\approx 0.70$  g, measured). No segment exceeds the plant's grip there; on the decaying surface the dotted line falls through the solid one.

**Table B1** Peak friction recovered from a synthetic brush-tire plant (true  $\mu = 0.70$ ), against excitation.

| $\delta$ amplitude | Grip reached | Bound $\hat{\mu}$ | Brush fit Eq. (16) |
|--------------------|--------------|-------------------|--------------------|
| 0.033 rad          | 45 %         | 0.312             | gated off          |
| 0.047 rad          | 63 %         | 0.442             | <b>0.699</b>       |
| 0.060 rad          | 80 %         | 0.557             | <b>0.699</b>       |
| 0.090 rad          | 100 %        | 0.700             | <b>0.697</b>       |

Column 2 is the fraction of the available grip the manoeuvre reached,  $\max |F_y|/(\mu F_z)$ ; column 3 is the utilisation bound  $\max |F_y|/F_z$ , which is what that fraction leaves as a lower bound on  $\mu$ . The brush fit of Eq. (16) instead recovers the peak from the curvature of  $F_y(\alpha)$  and is therefore flat in excitation.

The first row is refused by the absolute gate  $\max |F_y|/F_z \geq 0.40$  of Section 5.5.2: at  $\mu = 0.70$ , 45 % utilisation is only 0.31 of  $F_z$ . The utilisation bound is reported instead.

reconstructed  $I_z \dot{\omega}$  term and a curve whose sustained demand is only about 0.70 of the axle's peak force, and it dominates the identified peak factor's error on both surfaces. The decaying surface adds a lag rather than a new error term.

## Appendix C Attribution by Offline Replay

This appendix attributes the closed-loop ordering of Section 6.3 to individual settings, at the cost of leaving the loop. Every number below is an *offline replay* of the identification stage on the five 30 s buffers each arm collected during its own run, scored with the quantity the closed-loop tire-force benchmark itself reports, so these numbers and Table 5's are on the same scale. Section 6.8 summarises what follows.

### Damping the co-identification loop

Table C2 sweeps  $\beta$  of Eq. (14) at six iterations throughout; two columns are the argument. At  $\beta = 1.00$ , the inherited undamped update, axle force RMSE is 70.4 and 62.3 N, reached by degrading monotonically from 38.9 and 26.3 N at a single iteration: the loop amplifies its own extrapolation instead of refining the estimate. At  $\beta = 0.00$  the coefficients never move from the pinned prior, which is no identification at all, and that is the worst column, 92.5 and 71.8 N. The minimum sits between them at  $\beta = 0.20$ , 28.0 and 19.4 N, a mean of 23.7 N, and the identified  $BCD$  rises with  $\beta$  over the whole damped range (7.82 at  $\beta = 0$  to 12.75 at  $\beta = 0.50$ ; the undamped 12.42 is no longer ordered by it), so the damping trades how much of each fit is accepted against how much extrapolation error comes with it. Six iterations are therefore defensible on this evidence, provided the update is damped.

### The other settings

Pinning the peak factor accounts for most of what remains, and the decaying surface is where its value is easiest to state: an estimator that cannot move  $D$  has nothing to publish when the surface changes. Replayed on five buffers of a  $\mu = 1.05$  run, letting the fit move  $D$  gives a mean  $|D - \mu|$  of 0.585 and a front-stiffness spread of 32.9 kN/rad; pinning only the axles whose brush fit was released reduces those to 0.191 and 32.3,

**Table C2** Relaxation sweep. Five recorded buffers, six iterations, offline replay.

| $\beta$              | 0.00 | 0.05 | 0.10 | 0.15  | <b>0.20</b> | 0.25  | 0.30  | 0.50  | 1.00  |
|----------------------|------|------|------|-------|-------------|-------|-------|-------|-------|
| Front $F_y$ RMSE [N] | 92.5 | 68.3 | 48.6 | 33.9  | <b>28.0</b> | 31.2  | 36.4  | 55.6  | 70.4  |
| Rear $F_y$ RMSE [N]  | 71.8 | 49.0 | 31.3 | 19.8  | <b>19.4</b> | 26.0  | 30.2  | 38.4  | 62.3  |
| Front $BCD$ [/rad]   | 7.82 | 8.82 | 9.66 | 10.36 | 10.88       | 11.32 | 11.65 | 12.75 | 12.42 |

$\beta = 1.00$  is the inherited undamped update and  $\beta = 0.00$  the pinned prior, i.e. no identification. 0.20 ships.

and also holding the axles with no released measurement, the shipped rule, to 0.084 and 18.5 kN/rad. What removes most of the error is refusing to let an unmeasured  $D$  be fitted at all. Normalising the physics-informed objective is necessary but not sufficient alone: it moves the loss shares from 99.9996% on the dynamics residual to 1.47%, yet applied without the damping it makes the force metric slightly *worse* (77.1 and 76.2 N against 70.4 and 62.3 N), because an objective that now fits the data supplies a sharper trajectory for the undamped loop to amplify. Two settings deserve a weaker statement than the closed-loop comparison alone would support: the S4D residual is worth 21% on the force metric at matched settings and a single iteration (35.3 N against the baseline MLP’s 44.7 N), an offline replay and not a closed-loop result; and the residual *objective* is a wash, **physics\_informed** and plain **mse** being indistinguishable at 23.7 N each once the normalisation and the damping are both in place. We keep it because it costs nothing and is the paper’s stated formulation, not because it buys accuracy. The fast- $\mu$  derate is likewise not separated: it is active only in **Excitation-S4D-Pinn**, and on the decaying surface it sat at the upper end of its [0.4, 1.2] clamp over the last cycles, so the reference re-clamp of Section 6.7 is carried by the published  $D$  and not by the derate.

## Appendix D Simulator–ROS 2 Interface

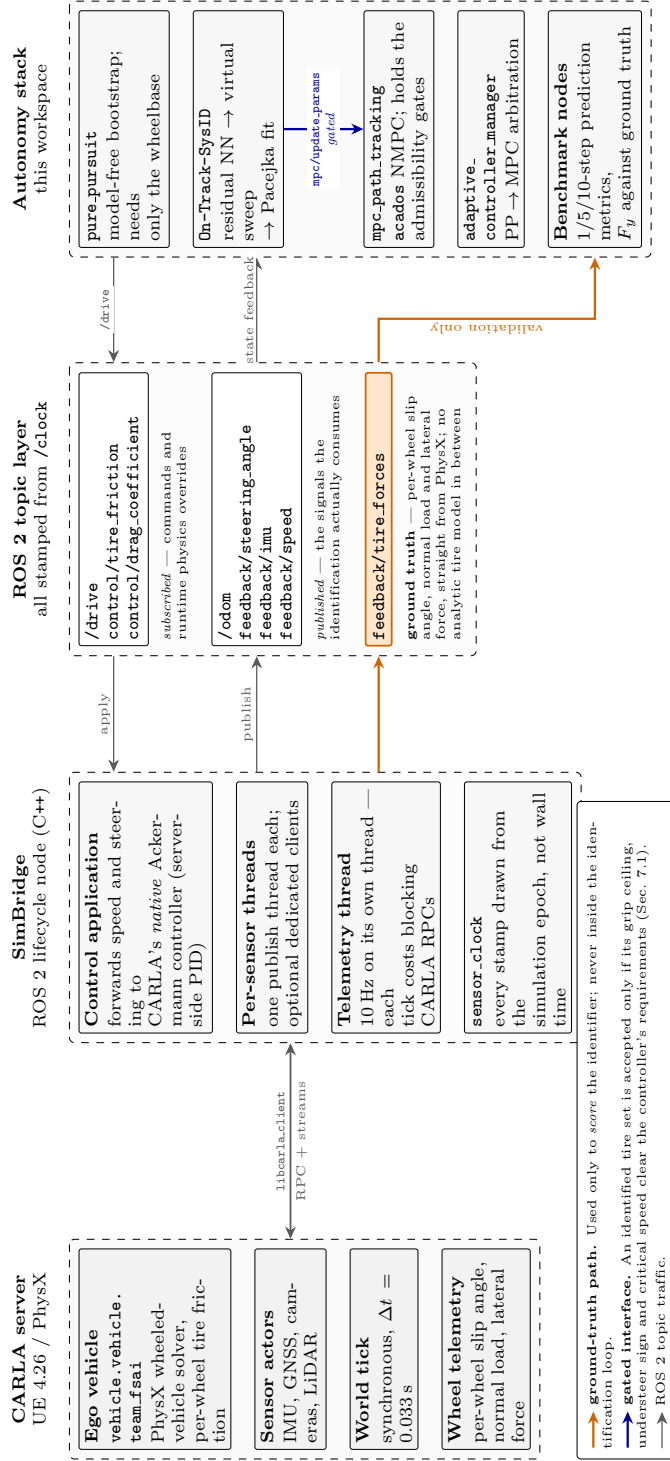
### Feedback path

Figure D2 shows the bridge and Table D3 lists the interface consumed by the identifier and its validation.

#### *Per-wheel ground truth*

**feedback/tire\_forces** carries CARLA’s own **WheelTelemetryData** in wheel order [FL, FR, RL, RR]. No analytic tire model exists inside the bridge, so nothing on this topic can disagree with the physics the vehicle is driven by. Four measured properties bound what may be read off it. (i) Only **lateral\_force** is force ground truth; it is the channel every validation in this paper uses. (ii) **longitudinal\_force** is a friction-capacity report rather than a measurement, tracking  $\mu F_z$  and correlating with  $ma_x$  at 0.04, and **wheel\_torque** is an exact identity of it. (iii) Below 0.5 m/s the publisher zeroes slip angle and both forces while the normal load stays live, so the identifier gates on the all-zero pattern as well. (iv) The slip angle needs no sign change across the boundary, verified rather than assumed.

Figure D3 states the conventions the rest of the paper assumes. CARLA inherits Unreal’s left-handed world while every ROS 2 topic follows REP-103 [50] and the

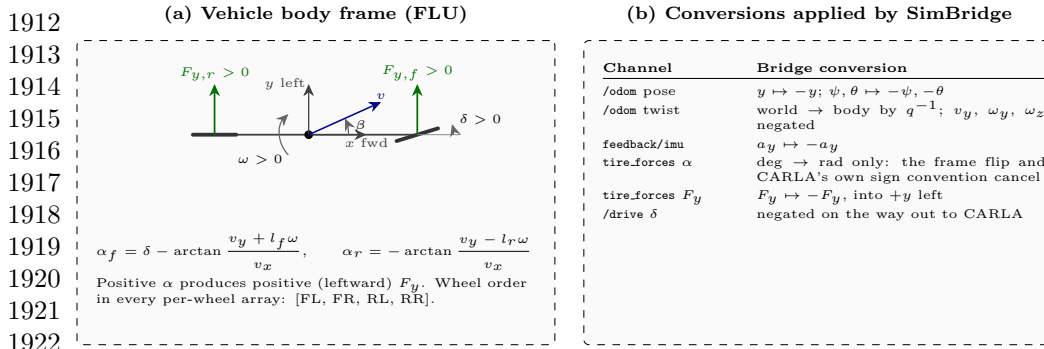


**Fig. D2** Architecture of the SimBridge ROS 2 interface. The bridge links against CARLA's C++ client, drives the ego vehicle through CARLA's native Ackermann controller, and republishes state, sensors and per-wheel telemetry on ROS 2 topics stamped from the simulation clock. The **feedback/tire\_forces** path (highlighted) carries the simulator's own per-wheel slip angle, normal load and lateral force, and is the ground truth against which the identifier is scored.

1887 **Table D3** ROS 2 interface used by the identification and validation pipeline. Namespace is `/sim`  
 1888 unless marked absolute.

| 1889 Topic                                 | Type                                              | Role                                                                                     |
|--------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------|
| 1890 <code>/odom</code>                    | <code>nav_msgs/Odometry</code>                    | State $v_x, v_y, \omega$ ; twist is body-frame                                           |
| 1891 <code>feedback/steering_angle</code>  | <code>std_msgs/Float32</code> (deg)               | Achieved road-wheel angle $\delta$                                                       |
| 1892 <code>feedback/imu</code>             | <code>sensor_msgs/Imu</code>                      | $a_x, a_y$ for the friction warm-start                                                   |
| 1893 <code>feedback/tire_forces</code>     | <code>sim_manager_msgs/TireForces</code>          | Per-wheel $\alpha$ , $F_z$ , $F_y$ , $\mu$ , $\omega_w$ , $\kappa$ : <b>ground truth</b> |
| 1894 <code>feedback/vehicle_physics</code> | <code>std_msgs/String</code> (JSON)               | Live wheel parameters of Eq. (8)                                                         |
| 1895 <code>feedback/speed</code>           | <code>std_msgs/Float32</code>                     | Forward speed echo                                                                       |
| 1896 <code>feedback/motors</code>          | <code>std_msgs/String</code> (JSON)               | Per-wheel speed/torque/brake                                                             |
| 1897 <code>/drive</code>                   | <code>ackermann_msgs/AckermannDriveStamped</code> | Control command in                                                                       |
| 1898 <code>control/tire_friction</code>    | <code>std_msgs/Float32</code>                     | Runtime grip override                                                                    |
| 1899 <code>/clock (abs.)</code>            | <code>roscpp_msgs/Clock</code>                    | Simulation time base                                                                     |

1902  
 1903 identification equations are written in the body frame with  $\alpha$  and  $F_y$  positive to the  
 1904 left, the SAE-style axis convention [4,51]. The bridge is a per-channel conversion rather  
 1905 than a single transform, and the trap is that the tire telemetry's slip angle needs *no*  
 1906 sign change, because the frame flip and CARLA's own convention cancel: an error  
 1907 there is invisible in the states and inverts the identified curve, so it was checked against  
 1908 a slip angle rebuilt from the chassis state in a steady 4 m/s turn, equal in magnitude  
 1909 to within 2 % front and 15 % rear.



1924 **Fig. D3** Frame and sign conventions across the simulator–ROS 2 boundary. (a) All identification  
 1925 quantities are defined in the body frame:  $x$  forward,  $y$  left,  $\omega$  and  $\delta$  positive counter-clockwise, and  
 1926 slip angle and lateral force positive to the left, so that a positive  $\alpha$  produces a positive  $F_y$  [51]. (b)  
 1927 The conversion applied per channel. The slip angle of the tire telemetry is the one exception that  
 1928 needs no sign change, because the frame flip and CARLA's own convention cancel.

## Appendix E Reference Implementation

The identification package, the ROS 2 bridge and the controller stack used in this paper are released as open source. ON-TRACK-SYSID extends the published baseline [6] (upstream: <https://github.com/ForzaETH/On-Track-SysID>) and is distributed under the MIT licence that upstream’s package manifest declares. SIMBRIDGE is a separate GPL-3.0 package, and the MPC and benchmarking packages ship with it, together with the scenario definitions, the exported result CSV files, and the figure and sweep generators. The links to SIMBRIDGE and to the project repository identify the authors and are therefore withheld from this blinded manuscript; they are on the title page and will be printed here in the accepted version.

**Supplementary information.** This article has accompanying supplementary information, submitted as a separate document (**supplementary.pdf**). It holds the residual model’s physics-augmented inputs and objective, the temporal residual and the exact reduction of its cost, the per-phase identification timing, the axis-by-axis comparison against the baseline pipeline as published, and the supporting result figures. No argument in this article depends on it.

## Declarations

- **Data Availability** Every figure and table in Section 6 is exported together with the CSV file holding its samples, and the cross-run comparison reads those CSV files rather than re-deriving any metric. The scenario definitions, the exported result CSV files, and the figure and sweep generators are released with the source repositories named in Appendix E. The repository links are withheld from this blinded manuscript and are on the title page.
- **Code Availability** The identification package, the simulator bridge, and the model predictive control and benchmarking packages are released as open source; see Appendix E. The identification package extends the released baseline of [6] under the MIT licence that its package manifest declares; the simulator bridge is GPL-3.0. The two run as separate ROS 2 overlays communicating only over ROS 2 topics and services, and neither links against the other.
- **Conflict of Interest** The authors declare no competing interests relevant to the content of this article.
- **Ethics approval** Not applicable. All results in this article are obtained in simulation. No experiment involved human subjects, personal data, live vertebrates, or the operation of a physical vehicle.
- **Funding** Not applicable. This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.
- **Author Contributions** Stated on the separate title page, as the journal’s double-blind procedure requires.